



Smart Connects: AI-Powered Platform for Connecting Individuals to Communities



By: Mohit Bhandari (NP069596)

NP3F2504IT

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Supervised by: Dr. Laxman Mandal

2nd Marker: Mr. Edward Basnet

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Yours sincerely ,

Mohit Bhandari

Abstract

The Smart Connects: AI-Powered platform for Connecting Individuals to Communities project seeks to tackle the difficulty individuals face when looking for and getting involved in communities which suit their needs (social, professional, academic, and personal) interests, and objectives. With so much information distributed amongst varying sources, from social networking profiles and groups to event pages, independent websites and much more, the discovery process can be difficult and very time consuming. To ease this challenge, it is proposed the development of an intelligent platform to connect willing individuals to active communities.

It enables users to register their interests, view communities and view events, whereas community leaders can set up a profile and advertise activities to the community to garner the interest of prospective members. Recommendation functions will be able to suggest communities to the user based on his or her interests. The goal of this system is to have a singular platform for communities to gather information on, and make it easy for members to participate in communities and build network relations

Keywords: Community Platform, AI Recommendation System, Community Engagement, Social Networking Platform, Event Discovery System

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Chapter 1: INTRODUCTION

1.1 Introduction

Communities play a pivotal role in fostering knowledge sharing, cooperation and interaction in today's digital world. There are many different kinds of communities like social clubs, professional groups, interest-based groups, voluntary associations etc. that provide us with ways to interact with others who have similar interests or objectives. Networking with others who have similar interests, personal and social development and social interactions can be provided by these communities and may be appealing to those who seek connections.

Although the number of both offline and online communities is increasing, finding appropriate ones based on the desired topic and interests remains difficult for many. This is particularly true of students and professionals who have recently moved to other cities/countries to attend school or start new jobs. These communities generally advertise information through social media profiles, messaging applications, or individual Web pages making it difficult to efficiently and easily locate appropriate ones (Shu-Hsien Liao, 2021).

Existing platforms such as Facebook Groups allow communities to create and manage groups within them. However users have to manually find groups that fit their interests; this becomes very time consuming to find the correct communities, they do not always provide intelligently recommends communities based on users behavior and interests and consequently users are not able to easily find the appropriate communities; Many communities on the other hand cannot find more and more participants interested to be a part of them (Nil-Jana Akpinar, 2025).

In order to solve the previously presented problems, Smart Connects: AI-Powered Platform for Connecting Individuals to Communities was introduced as an all-in-one platform, where individuals could easily find their communities and events that correspond to their interests, thus connecting people that search for activities with communities that search for members with smart recommendation features and a structured informative platform.

1.2 Problem Background

In the present digital era, communities and social groups play a significant role in providing a platform where individuals could communicate, share information and participate in social and professional activities. There are numerous communities such as clubs, volunteer organizations, student groups, and groups that meet to talk about hobbies, that enables individuals to build a relationship, share their knowledge, enhance their professional or personal skills (Paolo Spagnoletti, 2022) . Despite the plethora of communities that exist today, most people find it extremely challenging to identify communities that suit their interest, requirements and their location.

The main reason for this problem lies in the distribution of information about communities over a wide range of platforms like social media page, messengers, websites, event pages, and hence an individual needs to go through all these information manually to discover new communities and events which are of interest to them (Ellison et al., 2021). Such effort could be very tedious and may cause individuals, especially those new to a new city or country, to miss out on new and relevant communities/events.

In present, we have communities platform like Facebook Groups, other community platform to organize and publish their information. These platform rely on manual searching, category browsing and often does not provide results according to individual interest. Mostly users will find a number of unrelated groups, it does not necessarily means that this community fits with individual needs. While relatively smaller or new community will face problems in being found (Gediminas Adomavicius, 2021).

Also, in the absence of smart recommendation in many existing platform makes the process of finding a suitable community less efficient. Since it does not intelligently provide the communities according to user interest, it does not help individuals to find related communities while at the same time unable to provide sufficient exposure to communities to get new members. In this condition, there is a need of one place and intelligent platform which can gather the information about the communities and provides exposure for communities, at the same time helps people to find communities fitting with their needs.

1.3 Project Aim

The objective of this project is to build a smart connects platform an AI powered platform which aims at finding and connecting individuals with communities according to their interests. It aims at easing out the task of finding communities and events.

1.4 Objectives

- To create an centralized platform, where users can explore and connect with different communities and groups.

- To allow users to register and specify their interests in order to get relevant community suggestions.
- To let community organizers to create profiles and upload information about their communities and upcoming events.
- To improve community visibility and participation by providing an easy and user-friendly platform for engagement.

1.5 Scope

1.5.1 Tasks to be Executed

- **User Registration and Profile Setup:** User registration, profiling and declaration of interests allowing personalized community recommendations.
- **Community Management:** Community organizer registration and profile creation with detailed community information (activities, goals, contacts).
- **Community and Event Listing:** Provide a structured platform where communities can post information about their groups and upcoming events for users to explore.
- **Search and Recommendation Features:** Platform to showcase the groups and their events to the user.
- **User Interface Design:** User search functionality and some AI-based recommendations to find the right communities.
- **User Feedback and Evaluation:** User feedback collection mechanism (surveys, usability testing) to determine the effectiveness and ease of use.

1.5.2 Constraints

- **Technology Limitations:** The recommended services will only be primitive because there wasn't enough time and technology for developing the features in a full length of time during development process.
- **Availability Data:** The platform will not contain big external database but will use manually registered community at the beginning.
- **Development Resources:** The project development will be made under a certain amount of time which is available during the final year's academic project.

1.5.3 What Will Be Done

- Design and build a platform linking users to communities according to their interests.
- Include user registration, community profiles and community browsing function.
- Include event listings and community details for higher user interaction.
- Perform basic user testing for system and usability evaluation.

1.5.4 What Will Not Be Done

- No full integration will be done on this stage with external social media applications or any instant messenger.
- The sophisticated machine learning models to predict complex behavior will not be developed in this first version.
- Development of the mobile application will not be part of the project, web-based system will be the focus.

1.6 Potential Benefit

1.6.1 Tangible Benefit

- A user friendly way of finding relevant communities based on user needs.
- Community info and events could be searched quickly on a single platform.
- Community organizers could promote communities and events to a larger number of users.
- Participation rates are boosted through effective matching of users to the best communities.
- Users are saved from searching through different platform for community related information, with this organized system.
- Communities would increase their reach with a more enhanced discoverability.
- Event information could be accessed at ease, which will be the main concern for participants.

1.6.2 Intangible Benefit

- Social interaction between similar minded people will become much more intense.
- It enables and offers a good deal of personal growth and networking by joining in to the communities.
- The system helps a new student or professional to the city to adapt better to the city.
- Community is better connected to members and community relationship becomes more active.
- The system helps in teamwork, knowledge sharing and socialization between people.
- The system helps to become an active, socially connected community.

1.7 Target User

1.7.1 *Individuals New to a City or Country*

Students or professionals that are newly immigrated or moved to a new city/country struggle to know which groups or social circles they are part of. With the application, students or professionals can find the communities and events that best fit their needs, facilitate integration into the new social structure.

1.7.2 *Students*

Students will look for academic groups or hobby/volunteering groups or social circles. With the application, students will be able to easily find student clubs, study circles or social circles related to students' needs.

1.7.3 *Professionals and Job Seekers*

Professionals looking to broaden their network or gain new skills can leverage this platform to find professional communities, industry groups and networking events relevant to their professional lives.

1.7.4 *Community Organizers*

Community leaders, clubs, NGOs and social groups can utilize this platform to set up a page profile for their group/club, publicize events and attract new members.

1.7.5 *Older Adults and Elderly People*

Socially isolated older adults can connect with nearby groups, social clubs, or other activities through this platform that can help them develop relationships and participate in society. The

platform can facilitate to help them to be active, establish new friendships, and get involved with the community that contributes to enhance health and wellness.

1.7.6 Individuals Seeking Social Engagement

Individuals who seek involvement in social events, voluntary work, clubs dedicated to some special interest, or community activities can use this platform to discover these opportunities and communicate with other people having same interests.

1.8 Functionalities (Deliverables)

1.8.1 User Registration and Profile Creation

A user account can be set up in an easy way by entering minimum personal information and interests. These data will be used by system for recommending communities and events to user.

1.8.2 Community Discovery and Search

Users are able to discover and find communities in relation to category, i.e. Activities/hobbies, volunteering, professional and personal development, society or social networks etc. Information on the communities are made available in an orderly way.

1.8.3 Community Profiles and Information Pages

Each community should have a specific page where we present data like community description, activities, social media page, contact persons and some testimonials from users.

1.8.4 Event Listing and RSVP Feature

Communities can make known of any upcoming events, i.e. Users are to see what the events entail to and be able to confirm whether he/she will be attending by clicking the RSVP button.

1.8.5 Community Organizer Dashboard

Organizers can create, edit their community profile and event-plan and promote the event so that more and more members can be attracted to the community.

1.8.6 Admin Panel

Allows administrators to view and approve community registrations, manage users and guarantee that content is adequate and updated.

1.8.7 Feedback and Rating System

Allow users to vote and give comments on communities and events. Improves the trustworthiness of community content and supports the platform's evolution.

1.8.8 Basic Chatbot Assistance

Basic chatbot to help users discover communities or events according to their preferences or questions.

1.9 Alignment with the Sustainable Development Goals (SDGs)

SDG 11 Sustainable Cities and Communities aims to build inclusive safe and well-connected communities. The Smart Connects system addresses this SDG by giving people an easy-to-access place to find and interact with communities and events in their area which match their interests. This enhances the access to information available to people while simultaneously minimizing the barrier between individuals and community events.

The Smart Connects system will enhance the ability of users to be actively involved as the easy way to find communities and stay up to date about upcoming events makes users more aware and therefore increases community involvement. It acts to bring together individuals and improve social interaction between them. By also making it easier for community organizers to get the word out it also helps communities be more engaged.

In summary the Smart Connects system facilitates inclusion and the building of community through a streamlined and straightforward digital application. It will facilitate the building of integrated social environments and the active social engagement of users while strengthening connections.

1.10 Overview of the IR

This project proposes Smart Connects, an AI-powered system aimed at matching users to communities based on their interests, purposes, and social requirements. It aims at eliminating confusion caused by various disjointed information channels regarding communities and events and providing a one-stop shop and structured database for easily locating communities or events. This allows users to conveniently locate relevant

communities while at the same time providing opportunities for community organizers to post events and activities to their interested attendees.

This research investigates current systems and previous studies to identify deficiencies such as lack of personalization, inaccessible resources and scattered information; explores technical aspects of the system including tools, technology and system design; employs a structured development process supported by surveys to elicit user needs, pain points, interests, and preferences; identifies features like the ability to search for communities, view events, and get personalized community recommendations.

The interface of the system is planned to be user friendly and very simple to make it easy for various people regardless of their level of technical expertise to use it. In general the project would aim to create an efficient community discovering platform that will increase user engagement and facilitate better connections through a structured system.

1.10 Project Plan

1.10.1 *IR Project Timeline & Gantt Chart*

Table 1

Project Timeline for IR

Task	Start Date	End Date	Duration (days)	Chapter	Status
PPF	02/01/2025	02/08/2025	7	PPF	Completed
PSF	02/22/2025	03/01/2026	8	PSF	Completed
Ethic Form	02/22/2025	03/01/2026	8	Ethic Form	Completed

Chapter 1

1.1 Introduction	03/10/2026	03/14/2026	4	Ch1	Completed
1.2 Problem Background	03/14/2026	03/17/2026	3	Ch1	Completed
1.3 Project Aim	03/17/2026	03/18/2026	1	Ch1	Completed
1.4 Objectives	03/18/2026	03/20/2026	2	Ch1	Completed
1.5 Scope	03/20/2026	03/21/2026	1	Ch1	Completed
1.6 Potential Benefits	03/21/2026	03/23/2026	2	Ch1	Completed
1.7 Target User	03/23/2026	03/24/2026	1	Ch1	Completed
1.8 Functionalities (Deliverables)	03/24/2026	03/27/2026	3	Ch1	Completed
1.9 Overview of the Study	03/27/2026	03/28/2026	1	Ch1	Completed
1.10 Alignment with SDGs	03/28/2026	03/29/2026	1	Ch1	Completed
1.11 Project Plan	03/29/2026	03/30/2026	1	Ch1	Completed

Chapter 2

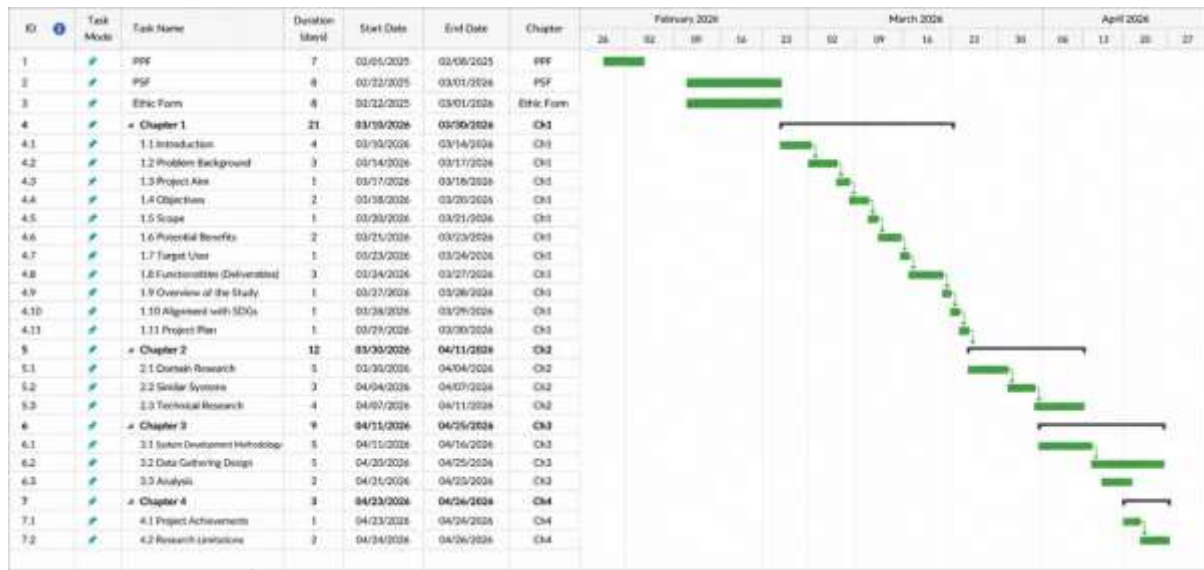
2.1 Domain Research	03/30/2026	04/04/2026	5	Ch2	Completed
2.2 Similar Systems	04/04/2026	04/07/2026	3	Ch2	Completed
2.3 Technical Research	04/07/2026	04/11/2026	4	Ch2	Completed

Chapter 3

3.1 System Development Methodology	04/11/2026	04/16/2026	5	Ch3	Completed
3.2 Data Gathering Design	04/20/2026	04/25/2026	5	Ch3	Completed
3.3 Analysis	04/21/2026	04/23/2026	2	Ch3	Completed

Chapter 4

4.1 Project Achievements	04/23/2026	04/24/2026	1	Ch4	Completed
4.2 Research Limitations	04/24/2026	04/26/2026	2	Ch4	Completed

Figure 1*IR Gantt Chart*

1.10.2 1 Year Project Timeline & Gantt Chart

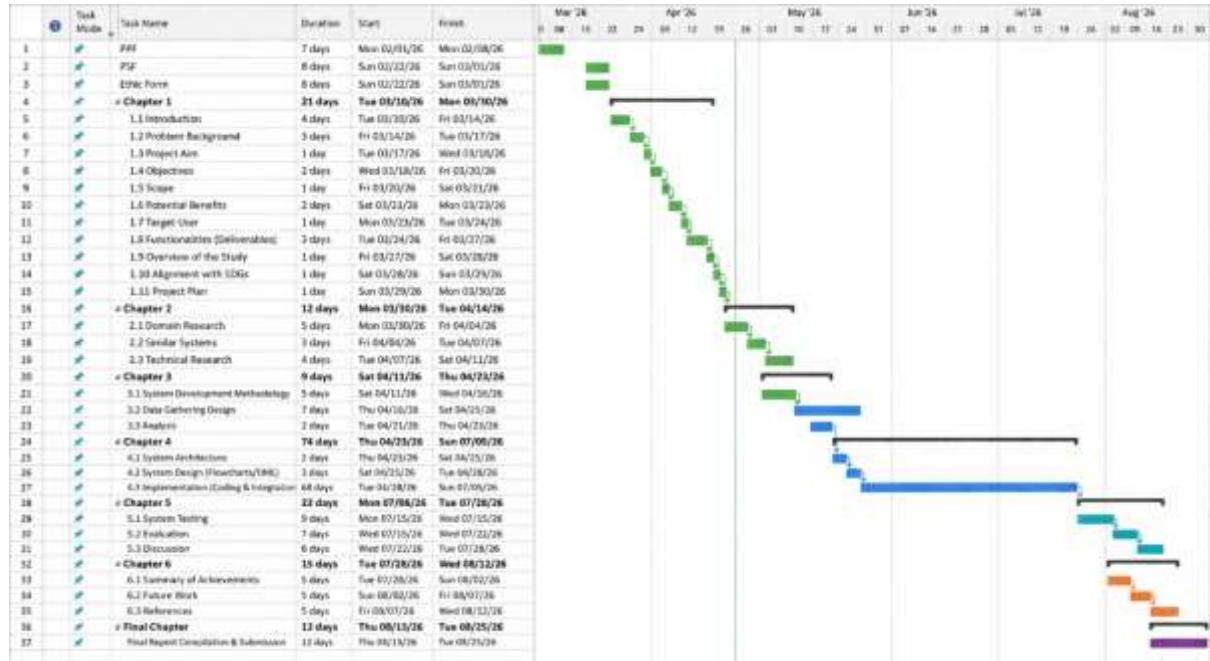
Task	Start Date	End Date	Duration	Chapter	Status
(days)					
PPF	02/01/2025	02/08/2025	7	PPF	Completed
PSF	02/22/2025	03/01/2026	8	PSF	Completed
Ethic Form	02/22/2025	03/01/2026	8	Ethic Form	Completed
Chapter 1					
1.1 Introduction	03/10/2026	03/14/2026	4	Ch1	Completed
1.2 Problem Background	03/14/2026	03/17/2026	3	Ch1	Completed

1.3 Project Aim	03/17/2026	03/18/2026	1	Ch1	Completed
1.4 Objectives	03/18/2026	03/20/2026	2	Ch1	Completed
1.5 Scope	03/20/2026	03/21/2026	1	Ch1	Completed
1.6 Potential Benefits	03/21/2026	03/23/2026	2	Ch1	Completed
1.7 Target User	03/23/2026	03/24/2026	1	Ch1	Completed
1.8 Functionalities (Deliverables)	03/24/2026	03/27/2026	3	Ch1	Completed
1.9 Overview of the Study	03/27/2026	03/28/2026	1	Ch1	Completed
1.10 Alignment with SDGs	03/28/2026	03/29/2026	1	Ch1	Completed
1.11 Project Plan	03/29/2026	03/30/2026	1	Ch1	Completed
Chapter 2					
2.1 Domain Research	03/30/2026	04/04/2026	5	Ch2	Completed
2.2 Similar Systems	04/04/2026	04/07/2026	3	Ch2	Completed
2.3 Technical Research	04/07/2026	04/11/2026	4	Ch2	Completed
Chapter 3					
3.1 System Development Methodology	04/11/2026	04/16/2026	5	Ch3	Completed
3.2 Data Gathering Design	04/16/2026	04/25/2026	5	Ch3	Completed
3.3 Analysis	04/21/2026	04/23/2026	2	Ch3	Completed
Chapter 4					
4.1 System Architecture	04/23/2026	04/25/2026	2	Ch4	In Progress

4.2 System Design (Flowcharts/UML)	04/25/2026	04/28/2026	3	Ch4	In Progress
4.3 Implementation (Coding & Integration)	04/28/2026	07/05/2026	68	Ch4	In Progress
Chapter 5					
5.1 System Testing	07/06/2026	07/15/2026	9	Ch5	Pending
5.2 Evaluation	07/15/2026	07/22/2026	7	Ch5	Pending
5.3 Discussion	07/22/2026	07/28/2026	6	Ch5	Pending
Chapter 6					
6.1 Summary of Achievements	07/28/2026	08/02/2026	5	Ch6	Pending
6.2 Future Work	08/02/2026	08/07/2026	5	Ch6	Pending
6.3 References	08/07/2026	08/12/2026	5	Ch6	Pending
Final Chapter					
Final Report Compilation & Submission	08/13/2026	08/25/2026	12	Final	Pending

Figure 2

1 Year Gantt Chart



Chapter 2: -LITERATURE REVIEW

2.1 Domain Research

2.1.1 *Community Recommendation Systems in Online Social Networks*

In this study the recommendation systems for finding relevant communities or groups within the online social network are investigated. It is describe how a recommendation system can find suitable communities for user by processing user's behavior, interests and relationships. Machine learning, data mining technology are utilized to process huge amount of data of users to find relationship or similarity between the users' interest. Collaborative filtering and content-based filtering were also discussed that these methods are commonly used for making recommendation system to finding communities matching to user profile (Kaur, 2020).

The primary goal of the study was to find a better approach to discover relevant communities from the online social networks that having tens of thousands communities or groups. Manual browsing through all of communities is inefficient and can be time consuming in large scale social network. So recommendation system is used to automatically recommend some relevant communities to users by considering users' profile. As the results shown, recommendation system with individual preference, historical activity on social network would highly increase the user engagement, and users can find a target community easily (Kaur, 2020).

2.1.2 *Online Community Platforms and Social Engagement*

In this research, it discusses how online community platform facilitates the social interactions and collaborative activities. It states that the web community is useful for people that share similar interests, objectives, professional and etc. From using a discussion board forum, networking platform or any other community-based web site, users are allowed to communicate and share information with others. This research states an effective community platform would facilitate to have more communication and enhance the user engagement (Molinillo et al., 2020).

The research sought to examine the effect of community variables on the moderating role of user engagement on the web. It indicates that user will be actively engaged with a community as long as they have a strong sense of bond and support in this community. This research proves that the users' engagement would be stronger if community has well organized and good way to achieve good communication channel, event and event activities and share goal, then user will have tendency to stick around a community in long term (Molinillo, July 2020).

2.1.3 *Artificial Intelligence in Social Networking Platforms*

In this research study, the utilization of artificial intelligence in enhancement of social networking and internet usage. This paper addresses how machine learning, natural language processing and sentiment analysis are utilized to determine users behavior and their likings; how artificial intelligence provides user customized experience; enhance user experience as

well as use information from the big amount of data produced from users and different social websites (Aichner, April 9, 2021).

The main objectives for this study were to examine how AI enhance the system of social networking system; it was discovered that social websites that have the application of artificial intelligence are more accurate in analyzing the user data and can provide better personalized services that will meet the user requirements; the study indicated that AI based system have learning capability as it analyze user behavior over a span of time, this makes it accurate in terms of recommendations and discovery of useful and relevant content or community (Aichner, April 9, 2021).

2.1.4 *Community Detection in Social Networks*

Community detection in social networks is a popular and growing area of research that is dedicated to finding groups of people who share common interest, behavior, or connections in a social network. A social network can be characterized by a vast amount of data arising from user activities in sharing and posting and any network activity performed by users. Algorithms or techniques that use the idea of network analysis are applied in the context of community detection in a social network. The underlying concept of community detection technique is to reveal structural properties in terms of relationships and associations between users. A significant number of studies attempt to identify groups or communities on the basis of connections of users or their shared interest (Santo Fortunato, 2022).

In this work, the aim is to further analyze how communities are spontaneously formed in a social network. Community detection algorithms allow network developers to identify

groups of users in order to support users to interact and receive relevant services on these platforms through recommendations systems which are expected to improve user engagement on social network sites. Based on the communities detected, recommendations systems will have the capacity to direct specific groups of users towards common interests, groups or conversations that suit them better. This study is valuable since it sheds light on the role community detection has for the improvement of Online Connection Platform (Santo Fortunato, 2022).

Table 2

Domain Research

Journal Name	Authors	Title	Key Findings	Research Gaps
Telematics and Informatics	Kaur, H., Paruthi, M., Islam, J., & Hollebeek, L. D. (2020)	Community Recommendation Systems in Online Social Networks	Recommendation systems using collaborative filtering and content-based filtering help users discover communities based on interests and behavioral patterns. These	Most existing systems focus on platform-specific recommendation s and do not support cross-platform community discovery or personalized matching across

			systems improve	different
			user engagement	networks.
			and participation	
			in online groups.	
Computers in	Molinillo,	Online	Digital	Many platforms
Human Behavior	S., Anaya-	Community	communities	focus on
	Sánchez,	Platforms and	improve	interaction
	R., &	Social	collaboration,	within existing
	Liébana-	Engagement	communication,	communities
	Cabanillas,		and user	rather than
	F. (2020)		engagement by	helping users
			connecting	discover new
			people with	communities
			similar interests.	based on
			Well-structured	interests or
			community	location.
			platforms	
			increase	
			participation and	
			strengthen social	
			relationships.	
Cyberpsychology	Aichner,	Artificial	AI technologies	AI integration in
, Behavior, and	T.,	Intelligence in	such as NLP and	community

Social	Grünfelder	Social	machine learning	discovery
Networking	, M., Maurer, O., & Jegeni, D. (2021)	Networking Platforms	help analyze user behavior and provide personalized recommendation s on social platforms, improving user experience and engagement.	systems is still limited, and many platforms do not fully utilize AI to recommend relevant communities or events.
Physics Reports	Fortunato, S., & Hric, D. (2016)	Community Detection in Social Networks	Community detection algorithms help identify clusters of users with similar interests in large networks, enabling better organization of social groups and improved	Many studies focus on theoretical algorithms but do not apply these methods directly to practical community discovery platforms for users.

recommendation
systems.

2.2 Similar Systems/ Works

2.2.1 Meetup

Link: <https://www.meetup.com/>

Figure 3

Meetup



Meetup is one popular Web platform, through which people are allowed to join local communities and attend local events in accordance with interests. For instance, members are allowed to view and find communities in interests like technology, leisure activity, job and networking, volunteer, social etc. And community organizers can create groups, organize meetings and invite group members to attend meetings. Meetup makes people easily get access to people who are with similar interests and it offers both online and offline community (Stephen Ricken, 2017).

Nevertheless, Meetup are focused on the events-based communities and the local gatherings; the recommendation mechanism is weak with a recommendation logic mostly dependent on locations and basic interest categories. User behavior and their personal characteristics play not strong part for an intelligent recommendation. The organizer's

advanced functions require paid subscriptions which make smaller communities hard to access.

2.2.2 *Circle*

Link: <https://circle.so/>

Figure 4

Circle.co



Circle.so is a management platform for communities that exist online, that creators, teachers, or groups can create to help them build and manage their communities privately. The platform helps organize discussions and content sharing within groups. With the tools available, community creators will be able to create groups where discussions can take place, posts can be shared and members can interact (yadav, 2026).

The Circle.so management tool helps community creators have a strong command over their communities, but it primarily focuses on existing and creator/organization-based communities; it does not guide users towards new communities and communities on different platforms. Users often sign up to communities either by invitations or direct links.

2.2.3 *Community*

Link: <https://www.community.com/>

Figure 5

Community.co



Community.com is a communication channel that allows businesses and people to communicate with their audience through messages. Community leaders, brands, and public people can interact with community members using text based communication. It enables businesses to manage their audience, send out announcements and notifications (Lima, 2025). But, Community.com centers on the interaction between businesses and audience instead of letting users discover communities or events. There is no place where users are able to discover and look for the right communities and events based on certain interests. Hence, it fails to facilitate community discovery or one-to-one communication.

Table 3*System Comparison Table*

Feature	Meetup	Circle.so	Community.com
Type of System	Event and community platform	Community management platform	Messaging community platform
Main Purpose	Discover events and groups	Manage private communities	Communicate with audiences
Community Discovery	Limited	Not available	Not available
Recommendation System	Basic	No	No
Community Creation	Yes	Yes	Yes
Event Management	Yes	Limited	No

Conclusion from Analysing Similar Systems

An examination of the current systems like Meetup, Circle.so and Community.com reveals that they can enhance and promote interaction and engagement in communities but they suffer from various shortcomings. Many of the communities are more or less dedicated towards either managing communities or organizing communities and do not focus much on

providing intelligent recommendations to the users about which communities are relevant to them based on their interests and preferences. There is a significant lacuna in terms of providing a centralized platform to connect users to communities across various categories and facilitating them with event discovery as well as community expansion. Smart Connects provides a solution to these problems by presenting a smart recommendation system powered by artificial intelligence to users and helping community organizers target their relevant user base.

2.3 Technical Research

2.3.1 System Requirement Analysis

Smart Connects system is a web based application which enable the users to identify communities/clubs based on interests, categories and location. The system need a moderate amount of processing power for database transactions, web server processing and user interface. The system requirements are divided into hardware and software requirement (McCabe, 2007).

Hardware Requirements

Table 4

Hardware Requirement

Component	Minimum Requirement	Justification
Processor (CPU)	Dual-core 2.0 GHz	Required to run development tools, database

		services, and local server environment efficiently.
Memory (RAM)	4 GB	Needed to run the web server, browser testing, and development tools simultaneously without performance issues.
Storage	1 GB free space	Storage is required for source code, database files, frameworks, and project resources.
Internet Connection	Broadband Internet	Required for research, accessing online libraries, downloading dependencies, and testing web deployment.
Input Devices	Keyboard and Mouse	Required for development activities such as coding, debugging, and system navigation.

Software Requirements

Table 5*Software Requirement*

Component	Requirement	Justification
Operating System	Windows 10/11	Windows is selected as the primary development environment because it supports the tools and technologies required for the project such as Node.js, PostgreSQL, and Visual Studio Code. It also provides stable compatibility with web development frameworks and testing environments.
Frontend Framework	Next.js	Next.js is used for building the frontend of the Smart Connects platform. It is a modern React framework that supports server-side rendering, efficient routing, and optimized performance

		which improves the overall user experience of the web application.
Backend Runtime	Node.js	Node.js is used to handle server-side operations of the platform. It provides a fast and scalable runtime environment that supports asynchronous processing and efficient handling of multiple user requests.
Backend Framework	Express.js	Express.js is a lightweight web framework used with Node.js to build RESTful APIs and manage routing, middleware, and server logic efficiently. It simplifies backend development and improves application structure.

Database Management System	PostgreSQL	PostgreSQL is used as the primary database system to store structured data such as user accounts, community profiles, events, and category information. It offers reliability, strong data integrity, and powerful query capabilities.
Web Server	Nginx	Nginx is used as a web server and reverse proxy to manage incoming requests and improve application performance. It helps handle traffic efficiently and ensures better scalability for the web platform.

2.3.2 *Programming Language Chosen*

1. Next.js

Figure 6*Next.js*

Next.js is the latest framework built on React used for web application front-end component development, it incorporates different sophisticated features like Server side rendering, automated routing and improved page load time which are very useful to reduce page load time and improve web user interface. The Next.js framework is being used to build the user interfaces for Smart Connects platform for several interactive pages such as home page, community pages, events page and user profile page (Stefan Tilkov, 2010).

Next.js also has the ability to both generate static and dynamic pages, this is very optimization friendly as the pages are search engine optimized and content delivered to the user in speed. Project structuring, re-usable components help simplify the development process. The above features of the framework helps us use it in building web interface for Smart Connects platform.

2. Express.js

Figure 7*Express.js*

Express.js is a minimalist, flexible and unopinionated web application framework for Node.js which is used for building web backend and API services. It helps to make building backend easier as it provides multiple utilities to route incoming request, to define and manage middle ware, handle requests from client and send the responses back to them. The Smart Connects platform is using Express.js to handle all its backend operations, this include user authentication, handling the search requests to the communities and interfacing with the PostgreSQL database, etc. (G, 2025).

Express.js provides the developers to structure their backend application into modular and organized manner. The system can also receive HTTP requests easily, providing good interaction between frontend and backend of the system. With extreme low weight, flexible and huge support of community for Node.js, it is a preferred choice for building web application.

Table 6

Comparison Table of Programming Languages

Criteria	JavaScript (Node.js)	Python	Java
Ease of Learning	Easy to learn with simple syntax and large developer community.	Easy syntax and beginner friendly.	More complex syntax and requires understanding of object-oriented concepts.

Web Development Support	Excellent support for web development with frameworks like Express.js and Next.js.	Supported with frameworks such as Django and Flask.	Supported through frameworks like Spring.
Performance	High performance with event-driven and asynchronous processing.	Moderate performance for web applications.	High performance but heavier runtime environment.
Library Ecosystem	Large ecosystem through Node Package Manager (NPM).	Strong ecosystem for data science and machine learning libraries.	Strong enterprise libraries but fewer lightweight web libraries.
Development Speed	Fast development due to reusable packages and flexible framework support.	Fast development for scripting and prototyping tasks.	Slower development due to more complex configuration.
Cross Platform Compatibility	Works on Windows, Linux, and macOS environments.	Cross-platform compatible.	Cross-platform compatible using

			Java Virtual Machine (JVM).
Suitability for the Project	Highly suitable for scalable web platforms and real- time applications such as Smart Connects.	More suitable for AI, machine learning, and data analysis tasks.	More suitable for enterprise-level systems rather than lightweight web platforms.

After this comparison JavaScript with Node.js is chosen as the main programming language. Node.js has been selected due to its great support to web application development, it is highly performing thanks to its asynchronous capability and a big ecosystem of libraries which can be used from NPM. It is easy to integrate JavaScript within framework like Next.js and Express.js, enabling fast development of the whole platform (frontend and backend).

2.3.3 Database Technology

1. PostgreSQL

Figure 8*PostgreSQL*

PostgreSQL is an advanced relational database management system which is free to use and open source; it can effectively store and manage data in a well-structured manner. PostgreSQL is reliable, scalable and also supports complex queries; this ensures that it is capable of producing sophisticated data management systems, useful for developers trying to deal with their specific data. Within the Smart Connects platform PostgreSQL would be ideal for storing and managing key information such as; user information, communities profiles, event information, category descriptions, and more.

Data integrity, performance optimized queries, and effectively indexed data make PostgreSQL a major asset for retrieving specific community and event information needed for a user search. Users' data are secure within the system and can be relied upon as protected. Performance, scalability, and stability are all advantages of using PostgreSQL that it is very well suited for creating web applications which require strong communities (Hasan Tolga Ünal, 2023).

2.3.4 Tools Chosen

1. Visual Studio Code (VSCode)

Figure 9*VSCode*

Visual Studio Code is a widely popular and strong source code editor application created by Microsoft. This IDE can be a very handy place for developers to write, edit and debug the application codes. There is support for many programming languages like JavaScript, HTML, CSS, SQL-this is a good set of languages that could be part of a web application project (Juan Pérez, 2021).

Some features that are present in this source code editor are syntax highlighting, debugging options, extensibility with plugins, integrated terminal, etc. It already supports a version control system like Git with the capability of managing changes in the code and teamwork. Other plugins are available for developers in the extension marketplace, which offer support for Node.js, management of databases and testing of APIs. With its lighter features and flexible design, Visual Studio Code seems to be a very suitable IDE to build the Smart Connects platform.

2.3.5 System Architecture of Smart Connects

The Smart Connects system has a system architecture based on modern web application architecture where frontend, backend, database and server are separated. The separated layers enable the system to scale, be maintained efficiently and allow easy communication among the layers of the system. The platform is a client-server architecture that users interact with a frontend interface that communicates with backend services to handle their requests, retrieve data from the database.

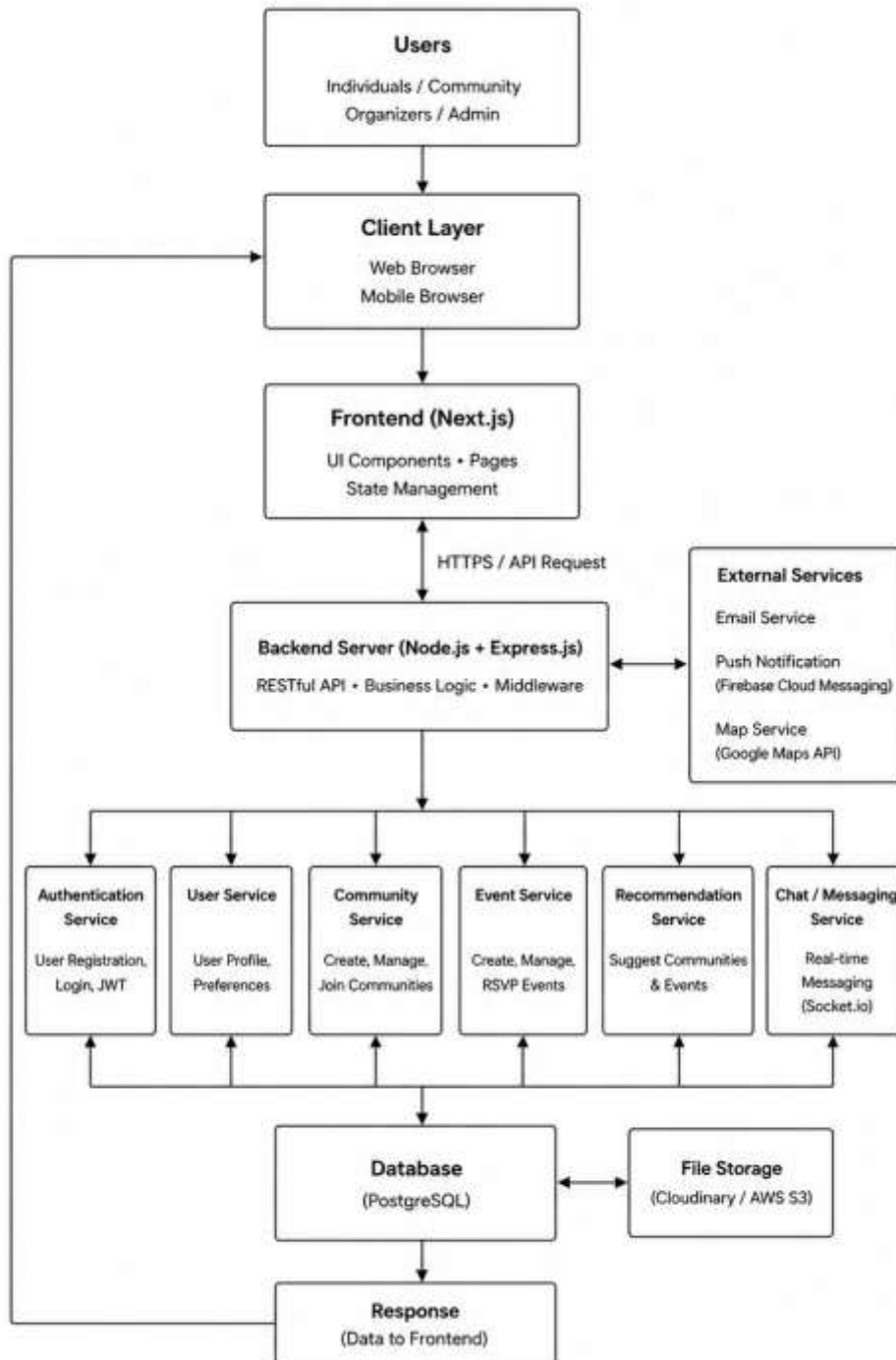
The frontend layer in the Smart Connects system is developed with Next.js that produces a responsive and interactive interface to interact with the users. This layer allows the users to sign up account, search communities, see event lists, browse community page and other tasks. Next.js manages the routing between pages, renders the components and connects with the backend with API calls. Next.js also increases the performance and loads pages faster.

The backend is developed using Node.js with the framework Express.js. The backend layer is the "backbone of the application", responsible for handling all the logic and requests received from the frontend. Its main concerns include user verification, handling community searches, event managing and data validation. The Express.js framework provides facilities for routing the requests to different endpoints and for developing mid-ware that are useful to help the application work smoothly. The communication between the frontend and the database can be handled by the backend efficiently and a lot clearer with the help of Express.js.

The database layer is maintained using PostgreSQL. It stores all the data for the application, such as user accounts, information about communities, the events on the platform, categories and keywords used for search. PostgreSQL ensures that all data is reliably stored and allows powerful queries needed to search effectively for the information about communities the user desires on the platform.

The server layer takes advantage of Nginx as a web server and reverse proxy in order to manage the traffic being sent to our application and render it effectively. System performance is enhanced as Nginx can control the amount of clients requesting the system and assist in load balancing as well as the communication to the client to ensure security.

All in all the Smart Connects system architecture enables all components to be well integrated. Next.js will serve as a development framework for the front-end, Node.js and Express.js as back-end handling, and PostgreSQL as our database and the Nginx web server in order to cater the application on the server side providing a robust and efficient architecture that links users with various communities and events.

Figure 10*Flowchart of Proposed System*

Chapter 3: METHODOLOGY

Methodology for development is the way in which the system can be planned, designed, developed and implemented. The development methodology ensures the project remains ordered and manageable and in context. The development methodology chosen for the Smart Connects project, considering the number of elements to be developed, from the Web based interface, backend services, database and the recommendation mechanism (Awed, 2025).

3.1 System Development Methodology

3.1.1 Introduction

Agile development with iterations is the process in which the Smart Connects platform is being developed. The agile development methodology creates software in small pieces of working functionality which are often referred to as iterations. Each iteration comprises requirement, analysis, design, development, test and review and the outcome is a working piece of the application that is then tested and refined in next iterations. This approach is appropriate for web based platform where functionality and user experience can change over time. In contrast to building the complete application, the Agile model permits the application designers to introduce individual features like user registration, community searching, events managing and recommendation engine at each iteration cycle and get feedback from users and supervisors (Samar Al-Saqqa, 2020).

3.1.2 Methodology Choice and Justification

The Smart Connects project uses the Agile development methodology as it fits well with features such as adaptability, improvement and incremental development. The platform

has a range of aspects such as frontend implementation, backend implementation (APIs, databases) and the recommendation features to take into account. With these developments it becomes possible to adjust the development of these systems to any changes/ improvements made within the development of Smart Connects by using shorter development cycles.

A benefit to using Agile development is the ability to react to any changes within the requirements. Within the development of Smart Connects the recommendations made by users/supervisors to add/remove specific elements such as; search filters, community features/recommendations, and User interface improvements can be easily adapted. Without having to impact the entire system structure there can be changes implemented for an improvement within the system.

An important factor to the Agile methodology is reducing risk factors within development. Developing the system in an incremental fashion would allow the use of core elements such as user authentication, community listing and database functions to be tested at the beginning stages of development. If any problems occur it will only impact small portions of the overall system making for easier fixability and making it more stable.

3.1.3 Development Phase Activities and Process

1. Planning Phase

The planning phase describes the general path that the Smart Connects platform should take. In this phase, the primary goal is to develop a web application that will link people to events and relevant communities. The main scope, functionalities, and system architecture were determined so that the development of the application would be able to proceed systematically.

Features are sorted into simple categories that aid in the managing of development. 'Must

Have' functionalities include user registration, user login, displaying communities and showing events. The filtering capabilities and other improvement functionalities were set aside to be incorporated later in order to present a simple, yet functional application. The required technologies such as Next.js on the frontend, Node.js and Express.js on the backend, and PostgreSQL for the database were all specified along with Nginx as the server application. Risks such as scarce data and time limitations were detected, and they will be taken into consideration throughout the project (Donatus Ebere Okonta, 2024).

2. System Requirement Analysis Phase

In this phase we clearly specify what is that the Smart Connects system will do and how will it function. In other words the idea of project is translated in to specific functional and non-functional requirements that should guide system design and implementation (Bashar Nuseibeh, 2019).

1. Functional Requirements

1. User Registration and Login

The system provides an interface for the user to create an account and login in a safe and secured environment. In the registration part, users enter basic information along with some interests. The interests are used in the system for customization of profile.

2. Community Listing

The platform will have different kind of communities like clubs, social groups etc to list in the system. The details of each community will be displayed including description, activities to know more about the communities.

3. Event Display

The system should display various kinds of events conducted for respective communities. Events details should also be listed like description, date, who all are participating and also an option to register for the event so the user will be able to know what is going to happen and can also register for the events.

4. Search Function

User should be able to search the community based on the given words or categories and the relevant results should be filtered in an organized way so that the time taken can be reduce.

5. Community Profile Management

Community organizers should be able to set up a profile for their community and can be updated. Community organizers should also be able to update the details along with events of their community so that they will also get the visibility on the platform.

2. Non-Functional Requirements

1. Performance

Pages should load quickly and the system must be responsive to user commands. A good performance rate helps the user to use the system effectively. Good performance would allow a good number of users to be online simultaneously.

2. Usability

The user interface has to be easy to use and simple. A user would not have difficulty navigating the application. An attractive design helps in the accessibility to all types of users.

3. **Security**

The system must be secured. Sensitive user data would be accessed through the user authentication and validation system. The unauthorized usage of a user account would be blocked. This helps to increase the user's trust.

4. **Scalability**

The system should not fail when there are a growing number of users and communities. A high number of data should still provide high performance. This helps to make the system a long-term application.

5. **Maintainability**

The application needs to be maintained easily, such that features are updated or fixed with the least impact to the entire application.

3. **Design Phase**

The design phase deals with defining the appearance and functioning of Smart Connects platform, i.e. Designing the UI, and defining data and user request processing. It transform the requirements into system structures so that it is easier to implement the system.

1. **User Interface Design**

The user interface is designed to be simple and easy to use for every user. It also have community category, featured communities, upcoming events section on homepage which makes it easier for the user to get to know things more readily. The clear navigation menu and clean interface of Smart Connects help the user in navigating through and knowing communities easily (Tom Černý, 2012).

2. Data Structure Design

The system has structured data which is stored in relational database. The system has used tables which help to store information such as account details of user, information about the communities, and information about events

3. System process Design

The mechanism is easy to work with request & response cycle. Any time the user searches or picks a community, the frontend send a request to the backend and the backend query the database to find related records. Finally, the backend returns the result to frontend for user to view clearly.

4. Testing and Evaluation Phase

Testing & Evaluation phase is important to verify that the Smart Connects platform is functional, usable and gives satisfactory performance. In this phase, testing of the system is carried out on various levels (Narula, 2026).

Types of Testing in the Project

1. System Testing

System Testing tests the overall features of the platform. System testing is to verify users can successfully register, log in to their account, listing all communities and displaying events. All system features will be tested.

2. Performance Testing

Performance testing will examine the response time and responsiveness of the

system. Page load time and user search/browse community speed will be tested to make sure it is fast and efficient enough.

3. User Acceptance Testing (UAT)

UAT will be done by users with using the platform to find bugs. Users search communities and check all events to have the system improved.

4. Unit and Integration Testing

Unit testing examines one single component which includes checking register and search function separately. Integration testing examines the communication between front-end, back-end and database to make sure the whole system work together.

This stage aims at finding out bug and improving the system response.

5. Maintenance Phase

Maintenance phase takes over to make sure the Smart Connects platform is up, running and secured and up to date post implementation. This phase includes fixing the system failures, tuning up system performance, incorporating the feedbacks (GeeksforGeeks, 2025).

1. Corrective Maintenance

Corrective maintenance takes care of the errors and bugs reported by users. For instance, if a user experience error in the event display, login or search feature then, the error will be addressed and corrected. In this process system performance can be improved and a system that is functioning smoothly and is good for the user can be developed.

2. Adaptive Maintenance

Adaptive maintenance helps the system function normally as the system itself is updated whenever required to suit in new environments. This type of maintenance is done when the system may have to be updated due to new frameworks, new database system or new server architecture and system remains working normal.

3. Perfective Maintenance

Perfective maintenance involves enhancing the performance of the system and to provide better user interface. It helps in making the system more user-friendly, more reliable, and also to improve the user interface by adding some features such as improvement in UI, addition of more features or searching speed, better navigations and others can be introduced by incorporating user suggestions.

4. Preventive Maintenance

Preventive maintenance aims at preventing any future errors or problems from the system and maintain a good system health by updating or improving the code and to check for security errors on the system so as to eliminate system errors.

3.1.4 Survey-Based Data Collection Design

The data collection method for this project chosen is surveys. This method is appropriate since users from different places would be included in the project and the data to be collected will have to be varied and diverse. Surveys can be used to collect both qualitative and quantitative data about user's behavior, needs and expectations toward finding and engaging within communities.

The consistency of survey answers and truthful response is assured due to the defined format of the question and also because they are provided anonymously. User's current

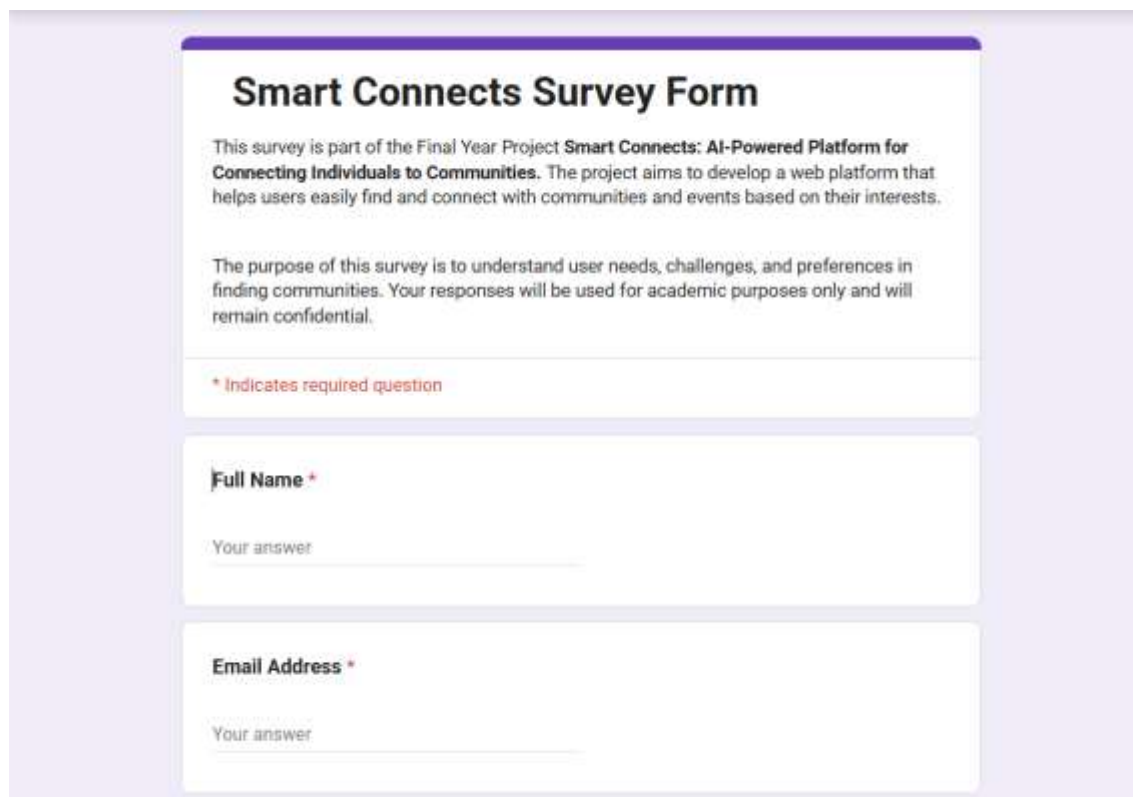
approach for finding communities and the pain points and desires the users might have is crucial information to develop such as a platform.

1. Section A: User Profile

Here, we have gathered some basic profile information from the users. It will provide insights about the user background and his/her online activities, as well as the way in which they participate in the online environment and their experience in using communities and events.

Figure 11

User Details



The image shows a survey form titled "Smart Connects Survey Form". The form is set against a light purple background. The title is in a bold, dark font. Below the title, there is a paragraph explaining that the survey is part of the Final Year Project "Smart Connects: AI-Powered Platform for Connecting Individuals to Communities" and aims to develop a web platform that helps users find and connect with communities and events based on their interests. Another paragraph states the purpose of the survey is to understand user needs, challenges, and preferences in finding communities, and that responses will be used for academic purposes only and remain confidential. A red asterisk indicates required questions. The form includes two input fields: "Full Name *" and "Email Address *", each with a "Your answer" placeholder text.

Smart Connects Survey Form

This survey is part of the Final Year Project **Smart Connects: AI-Powered Platform for Connecting Individuals to Communities**. The project aims to develop a web platform that helps users easily find and connect with communities and events based on their interests.

The purpose of this survey is to understand user needs, challenges, and preferences in finding communities. Your responses will be used for academic purposes only and will remain confidential.

* Indicates required question

Full Name *

Your answer

Email Address *

Your answer

Figure 12*User Online Behavior*

The image shows a survey form with four questions, each in a separate box. The questions are:

- 1. What best describes you? ***
 - ☐ Student
 - ☐ Working Professional
 - ☐ Business Owner
 - ☐ Other: _____
- 2. What is your age group? ***
 - ☐ Below 18
 - ☐ 18-24
 - ☐ 25-34
 - ☐ 35-44
 - ☐ 45-54
 - ☐ 55-64
 - ☐ 65 and above
- 3. How often do you use the internet? ***
 - ☐ Daily
 - ☐ Few times a week
 - ☐ Occasionally
- 4. Have you ever joined any community or group before? ***
 - ☐ Yes
 - ☐ No

2. Section B: Current Issues and Needs

This section aims to recognize user difficulties in searching for communities and events. It aims to comprehend user requirements in order to increase usability and convenience in finding communities with the Smart Connects system.

5. How do you currently find communities or events? *

☐ Social Media
☐ Friends/Family
☐ College
☐ Other: _____

6. How difficult is it for you to find the right community? *

1 2 3 4 5
 Very Easy ☐ ☐ ☐ ☐ ☐ Very Difficult

7. What problems do you face while finding communities? *

☐ Hard to find relevant groups
☐ Too many options
☐ Lack of information
☐ Not active communities

8. How often do you face these problems? *

☐ Always
☐ Sometimes
☐ Rarely

9. What type of communities are you interested in? *

☐ Social clubs
☐ Professional networks
☐ NGOs
☐ Hobby groups
☐ Sports and fitness
☐ Tech and IT
☐ Travel and adventure
☐ Women community networks (Aama Samuha)
☐ Senior citizen communities

10. Have you ever missed an event due to lack of information? *

☐ Yes
☐ No

3. Section C: Platform Preferences

This section aims to recognize user expectations on a community platform. It collects data on what kind of features user want in a community platform, and the overall usability of the system, also the interaction with the user interests and activities.

11. Would you like a platform that suggests communities based on your interests? *

☐ Yes

☐ No

12. What features would you prefer in such a platform? *

☐ Community search

☐ Event updates

☐ Recommendations

☐ Reviews and ratings

☐ Location-based community suggestions

☐ Easy membership/join option

13. How important is it for you to see upcoming events? *

☐ Very Important

☐ Important

☐ Not Important

14. How likely are you to join a new community through an online platform? *

☐ Very Likely

☐ Likely

☐ Not Important

☐ Not Likely

15. What device do you usually use to access online platforms? *

☐ Mobile

☐ Laptop/Desktop

☐ Both

4. Section D: Privacy, Customization and Feedback

This section aims to collect data on user worries concerning the data privacy and ownership of personal data, also allow users to offer suggestions, improvements, and feedbacks on the system.

16. Are you concerned about your personal data privacy? *

☐ Yes

☐ No

17. What privacy features would you prefer? *

☐ Control over data sharing

☐ Secure login

☐ Limited data collection

18. Would you prefer personalized recommendations based on your interests? *

☐ Yes

☐ No

19. What features or improvements would you like to see in Smart Connects to enhance community discovery and user experience? *

Your answer

I sincerely thank all participants who completed the Smart Connects survey. Your responses provided valuable insights for this project and will help improve the platform.

Thank you for your time and support.

Regards,
Mohit Bhandari

[Back](#) [Submit](#) [Clear form](#)

5. Survey Question Types

- Multiple-Choice Questions
- Multiple-Response Questions
- Frequency-Based Questions
- Open-Ended Questions
- Demographic Questions

The Smart Connects system applies a well-defined structure survey to identify user requirements and needs from various user categories including students, professionals and the public users. The system uses both quantitative and qualitative ways to gather user's needs and expectations and overcome issues.

The survey targets on how users discover their interested communities and events currently, the problem they encounters, and what they expect from the system. Those gathered data would improve the system design, usability, user's experience of using Smart Connects and etc. All the data collection would follow ethical consideration of collecting user data. The participants provide consent and their collected data would be protected in a private way and solely used for research (Creswell & J. David Creswell, 2018).

3.1.5 Data Gathering Techniques

Data Collection is a vital part in the system development process, which provides necessary knowledge to identify users' expectations, preferences and pain areas. Hence, in Smart Connects system, collection of the correct set of data is necessary to design the system which will be usable, effective, appropriate and in accordance to the real user needs. It helps making efficient and correct decisions in the design and development phase of the system.

In the development of this system, survey has been chosen as the most efficient method to obtain the required data. Survey method enables collection of both quantitative and qualitative data from the large population, in a structured form. The ease with which this method could be implemented with student, professionals and the people from general background who is interested in communities and events make this system be preferred for the data collection method (Ponto, 2015).

The survey has been categorized into different section like user profile, current problems, system preferences and privacy with comments section to obtain information based on users.

Advantages of Using Surveys

1. Effectiveness

Surveys enable to collect information from numerous users simultaneously which saves time to collect vast amount of information.

2. Consistency

Since it consists of questions with fixed options or choices that ensures all users answer in a similar manner, the data is easy to compare with one another and to analyze.

3. Quantitative Insights

Multiple choices, scale questions that give options to respond for how many times something happened to provide accurate and measurable data helping understand various patterns and trends.

4. Qualitative Insights

Open-ended questions in the survey help the users express their own opinions and provide suggestions, to help understand the user requirements to build a simple and easy-to-use platform.

5. User's Needs

Questions on problem faced or preferences help the users to express their needs, which support designing the simplest user friendly platform.

All gathered data is used throughout the development process to further enhance the system at each stage of development, user feedback and data is incorporated to refine existing functionality, usability is improved and the platform is designed in accordance to the real-world. This will further contribute to an ever improving Smart Connects platform.

Using the survey methodology gives a precise overview of user behavior, needs and expectations in discovering communities, in turn guiding the decisions taken during development to create an efficient, user friendly and need fulfilling platform.

3.2 Analysis

This project uses mixed methods approach for analyzing data which includes both qualitative and quantitative methods. Quantitative data obtained through survey methods has been analyzed by descriptive statistics using frequency and percentage distribution to identify patterns in user answers. This aids in identifying user behavior, usage preferences, common problems encountered by the users.

Qualitative data gathered through open-ended answers has been analyzed using methods such as coding in order to categorize responses, extract the most common themes and also to identify the common issues/feedback. The combined analysis helps to extract common views, user expectations and ideas for enhancing the system (Clarke, 2006).

3.2.1 Analysis Towards the Data Retrieve from The Data Gathering Activities

The analysis from the survey taken with 34 respondents is shown below. The intention of the survey was to get the main user, understand the difficulty users have in searching communities, and know the user expectations of Smart Connects platform. The survey is to find the need of the users and also to prove the problem addressed in the project.

It is observed that the primary user is students, followed by working professionals. 64.7% of the respondents are students and 26.5% are working professionals. This shows that the users would predominantly be young people with good networking skills and eagerness to learn and communicate. Regarding age, majority users are in the 18-24 range, followed by 25-34 range. This represents young digitally active users. It has also been revealed that most of the users are actively connected to the internet with 91.2% of the respondents using the internet daily. As of today, 70.6% of the respondents are currently using social media to search communities, 11.8% are searching from their college websites, and 17.6% are relying on their friends. This clearly shows that there is no centralized community searching platform available for the users.

The data has also clearly demonstrated the presence of the problem addressed by the project. As many as 52.9% users struggle with finding communities and others reported issues such as lack of information(41.2%), the options available are too many (32.4%), the community is not very active(23.5%). Out of all the respondents, 67.6% are regularly facing these problems. It has also been proved that 79.4% of users have missed an event due to lack of information. The respondents have a clear interest in clubs and professional networks, 64.7% and 55.9% respectively.

The users expect Smart Connects to have a personalized and smart design. The users strongly prefer the community to suggest the clubs based on the user's interests and 91.2% preferred if clubs could be suggested based on user interest. 94.1% are interested in getting personal recommendations and key features users look for include event updates(70.6%), search with respect to location (61.8%), easy membership processes. It seems the user wants their experience with Smart Connects to be intuitive and effortless.

The user requires the system to be responsive and compatible with both mobile and desktop platforms, as 50% of the respondents access the internet using both types of devices. Privacy is also of great concern, as 97.1% respondents care about how their data is being secured. To be specific, they desire a mechanism that allows users to share their data when needed and also a secure login mechanism for their account. The open ended responses further indicated the user's interest in enhanced search functions, location-based suggestions, and real time updates in their app. They also stressed upon information accuracy, privacy controls and better communication features.

Hence, the data clearly depicts that users face a regular problem in searching communities and need to have better ways for accessing event related data. They would be more interested in a smart and interactive platform which offers personalization, security and seamless community discovery.

3.2.2 Final List of User Requirement Summarized from The Analysis

Figure 13

Full Name

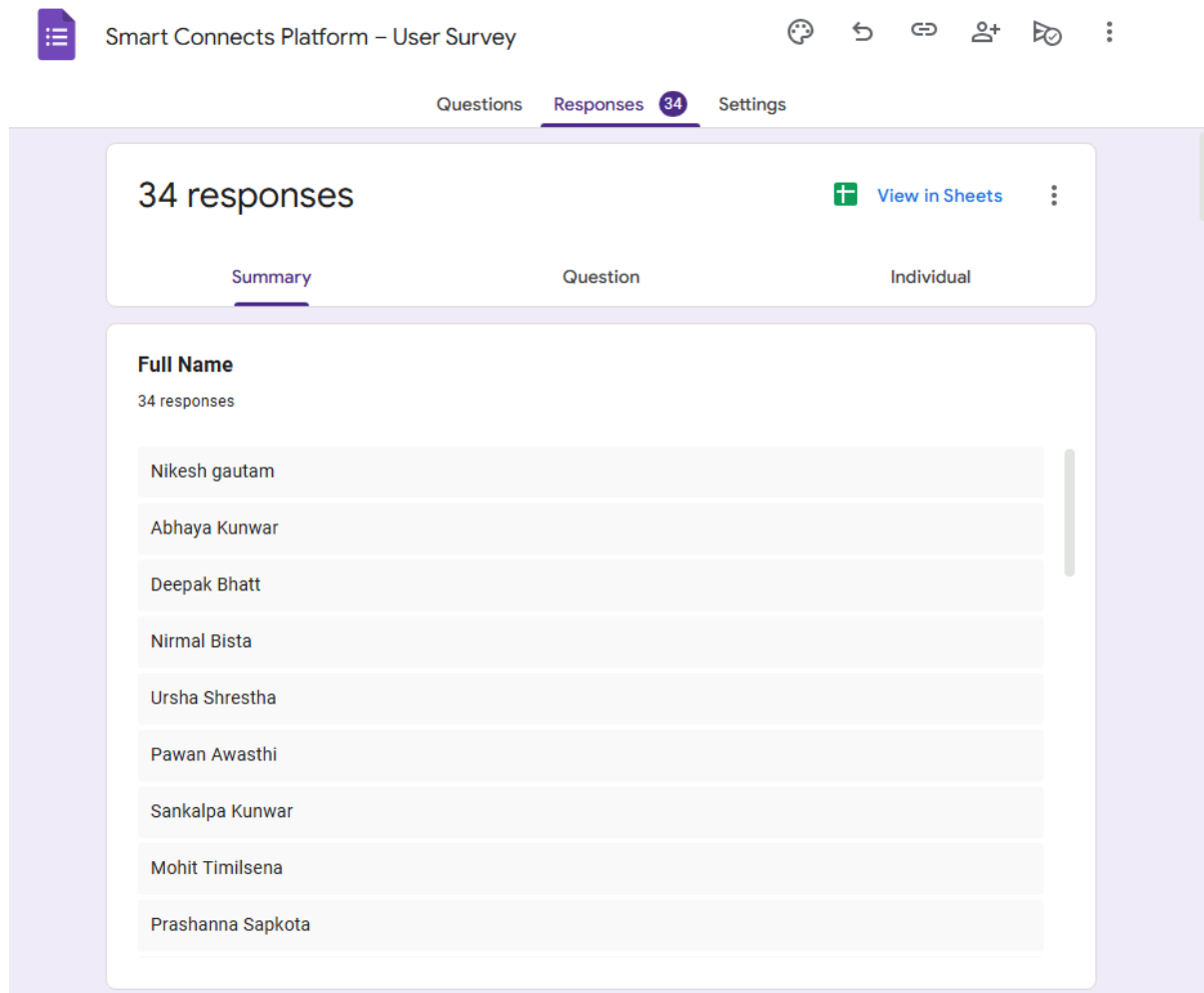
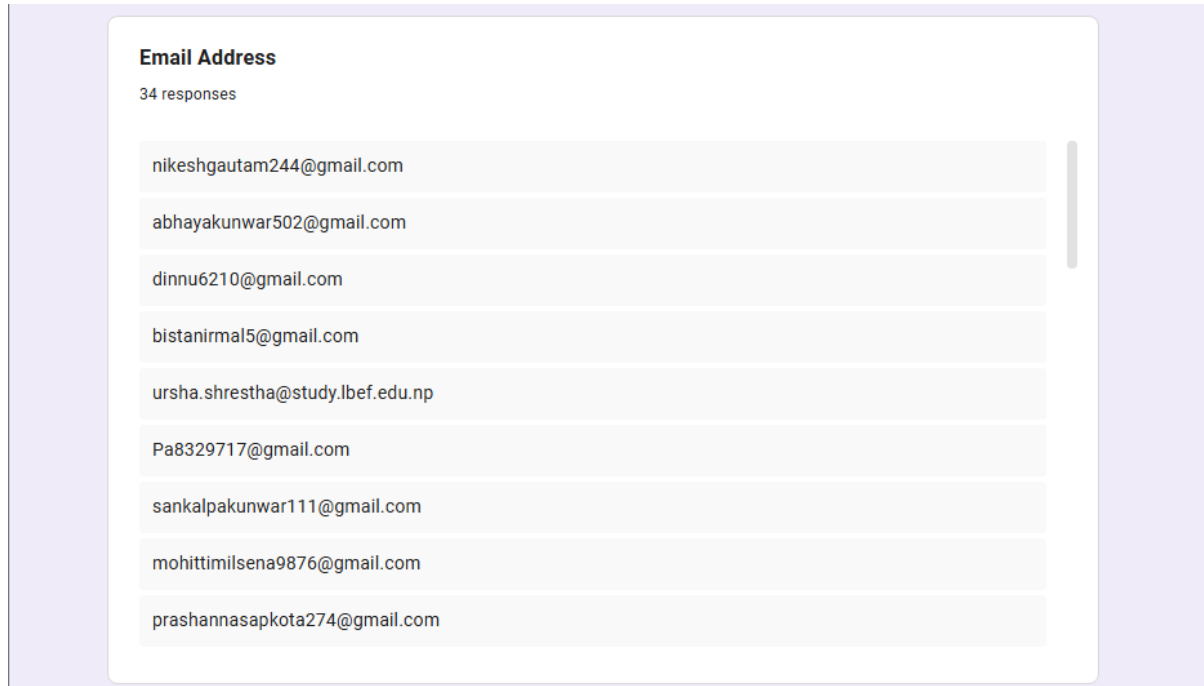


Figure 14*Email Address***Figure 15***Question No. 1***1. What best describes you?**

34 responses

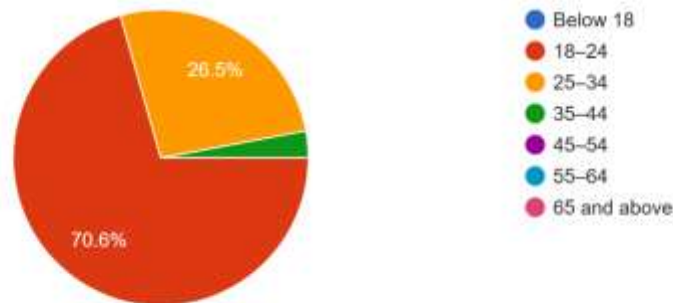


As per the survey results, it is clearly stated that the most of the users are students, representing 64.7% of the total and are the main target users for Smart Connects platform. While 26.5% of the users are working professionals, which are the strong second category of users, few business owners or combination of students and workers. The platform mainly should serve for the students while also cater for the working professionals for better usage.

Figure 16

Question No. 2

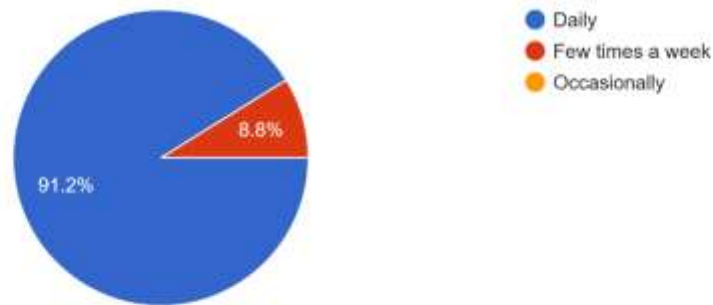
2. What is your age group?
34 responses



The survey shows that the respondents fall in to the age group of 18-24 in large numbers, representing 70.6%. Followed by 25-34 which is 26.5% and few in the 35-44 age group. This indicates that the target users are young ones mostly the students and professionals. These are the frequent online users and have active participation in the platforms and activities happening online.

Figure 17*Question No. 3*

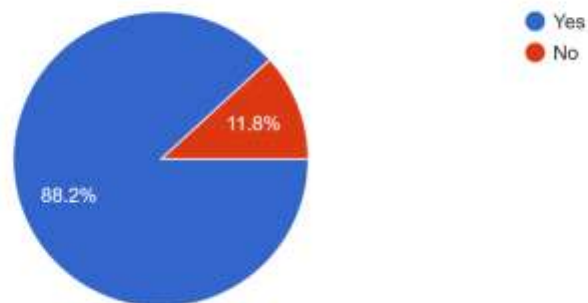
3. How often do you use the internet?
34 responses



The survey shows that the users are daily internet users (91.2%). The few of them uses few times a week and no users use occasional internet. This indicates that the users are online frequently. This information proves the necessity for a web based application for the users.

Figure 18*Question No. 4*

4. Have you ever joined any community or group before?
34 responses

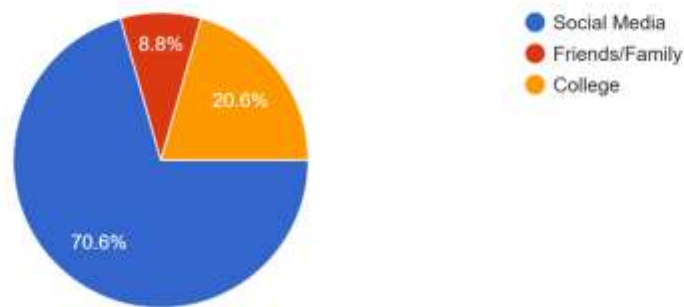


The survey shows that majority of the users are already in a community (88.2%). And very less users are not yet in any community (11.8%). This shows that most of the users already have experience of being a part of a community and will understand the value of the Smart Connects platform for finding communities easily.

Figure 19

Question No. 5

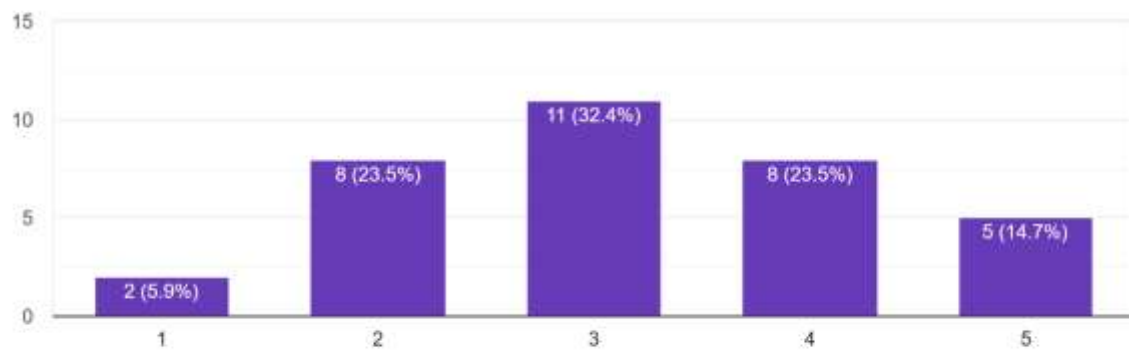
5. How do you currently find communities or events?
34 responses



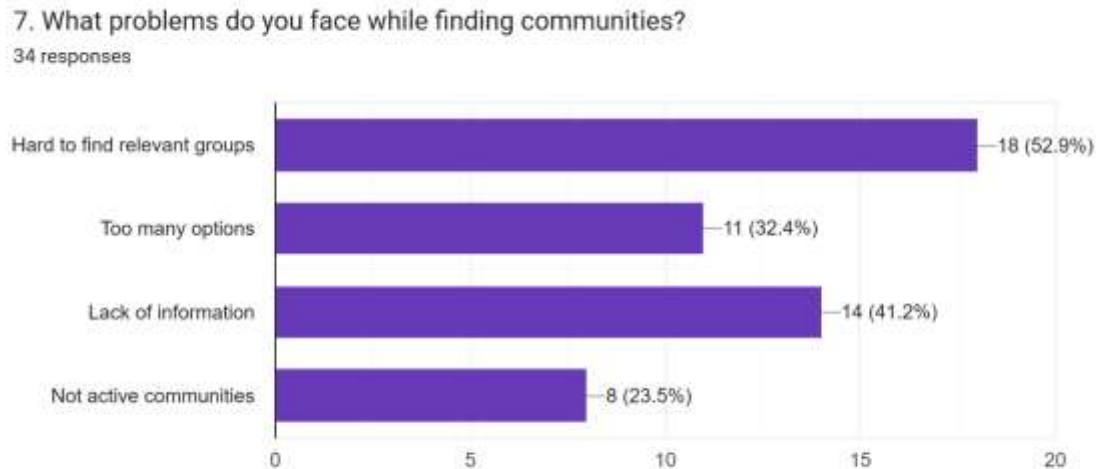
The survey shows that most of the users are aware of communities and events through the internet or social media (70.6%). Students have been informed through their colleges (20.6%) and their friends and family (8.8%). There is a significant need for a platform that can consolidate the way users find and discover communities, as they are using social media but many offline methods as well.

Figure 20*Question No. 6*

6. How difficult is it for you to find the right community?
34 responses



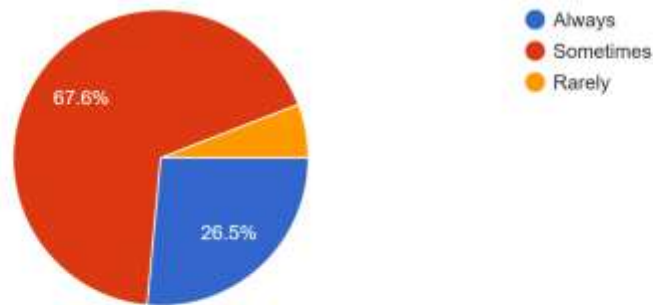
From the results, the most rating were given to 3. The finding that 32.4 % chose 3 in the above graph proves that most people felt the task was average. With 23.5 % rated 2 and 4 respectively and 14.7 % rated 5 it proves that the finding of a community is a noticeable challenge and many people face this problem in general. This proves that an effective platform is indeed required such as Smart Connects.

Figure 21*Question No. 7*

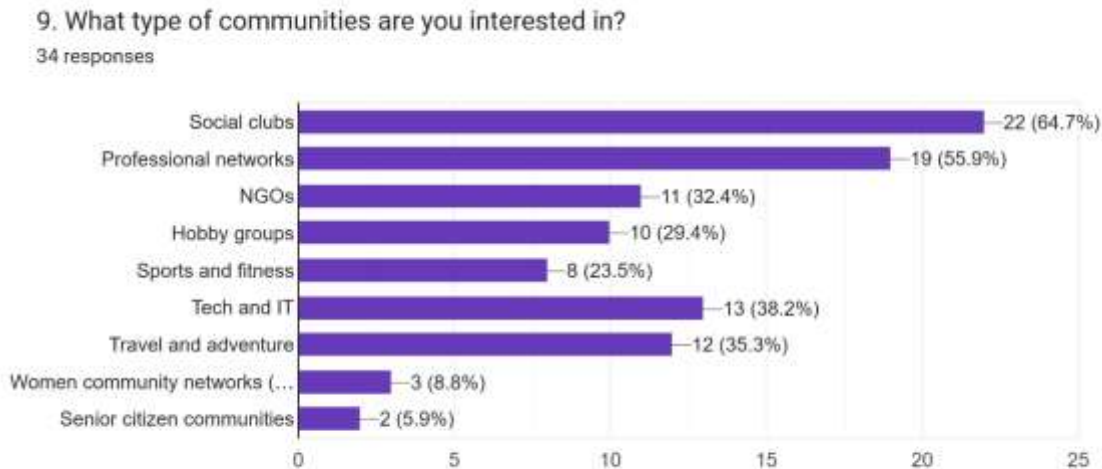
From the survey result it proves that the main problem is difficulty of finding the right communities, as most chose 52.9%. Other issues like less information (41.2 %), too many communities (32.4 %), Inactive Communities (23.5%). This shows that users found relevant issues with finding communities, lack of good content and inactive ones so an effective platform such as Smart Connects is indeed necessary.

Figure 22*Question No. 8*

8. How often do you face these problems?
34 responses



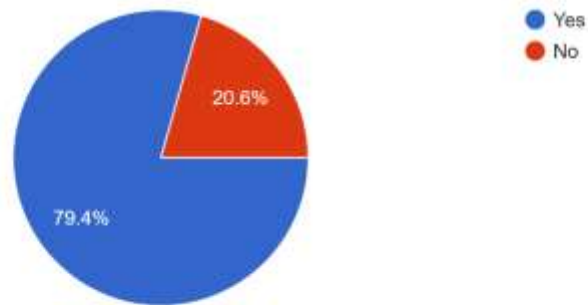
From the survey result most people feel they always have problems finding communities, as 26.5 % felt they always do, 67.6% sometimes do and a very little amount 5.9% rarely does. This proves the requirement of a better platform.

Figure 23*Question No. 9*

As is clearly seen from the graph the most preferable communities were for members of clubs (64.7%) and for professional organizations (55.9%). Furthermore the users showed interest in tech and IT (38.2%), travelling and adventure (35.3%) and NGOs (32.4%). Women and seniors interests and interests in hobby and sport and fitness clubs were less dominant. This demonstrated that people are mostly interested in social and professional communities and it should be the main priority for Smart Connects.

Figure 24*Question No. 10*

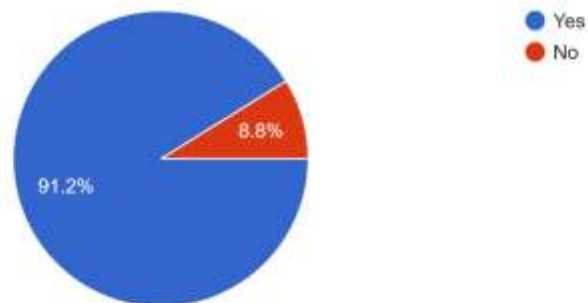
10. Have you ever missed an event due to lack of information?
34 responses



From the survey, it was proved that 79.4 % users have missed an event because of non availability of relevant information. 20.6% users have never faced the problem. Hence an effective application such as Smart Connects is definitely required.

Figure 25*Question No. 11*

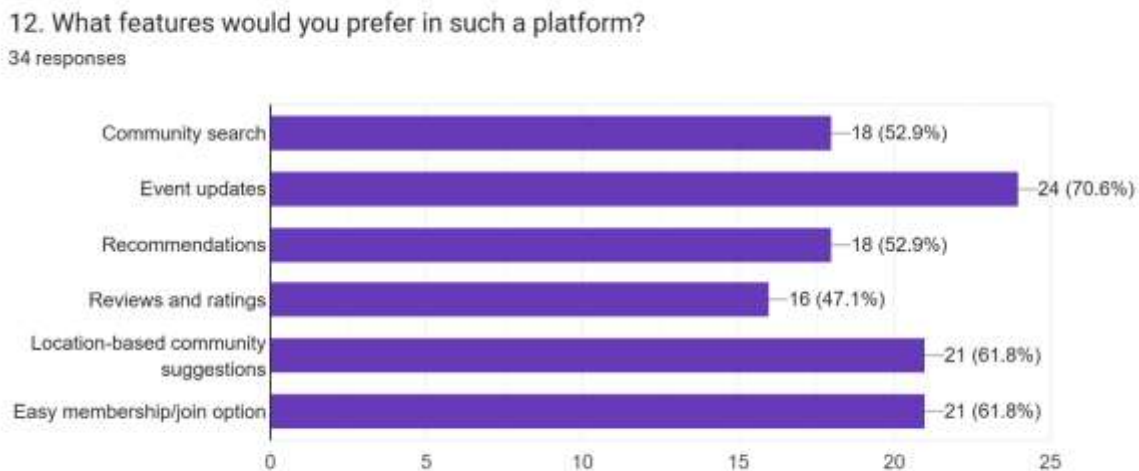
11. Would you like a platform that suggests communities based on your interests?
34 responses



The survey clearly shows that over 91% of people would want the platform to provide communities based on user's interest, whereas only a very small percent of people are not interested (8.8%). Therefore, personalized recommendation becomes one of the important features for Smart Connects.

Figure 26

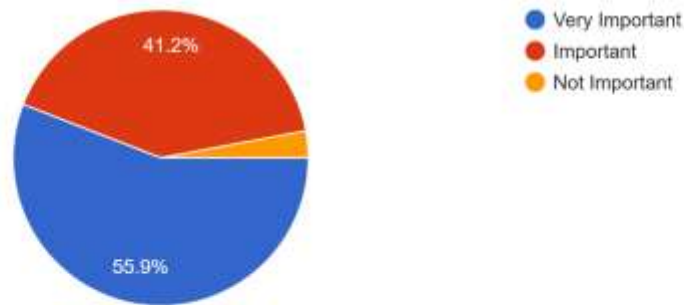
Question No. 12



The graph clearly shows that event updates is one of the most favorable features chosen by over 70% of people, followed by location suggestions, membership convenience and community search, suggestions of communities (respectively chosen by over 61% and 52%). Reviews and ratings was also a popular choice by users of Smart Connects with about 47% opting for it. This means that users require and demand timeliness and easy membership, easy suggestions, searching communities.

Figure 27*Question No. 13*

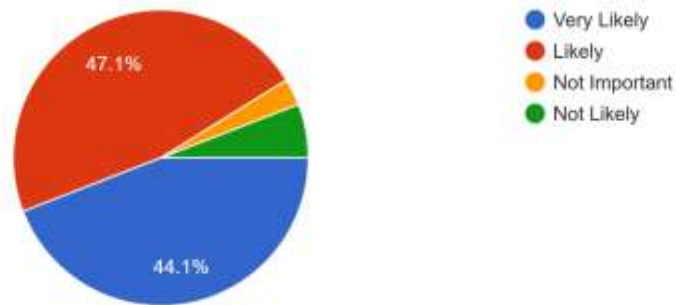
13. How important is it for you to see upcoming events?
34 responses



The survey results clearly shows that over 55% of users are interested to know what's going to happen next, while 41% is important. Only a very few percentage of users (nearly 3%) says they're not interested in it. This implies that people want to have a view on what's coming up.

Figure 28*Question No. 14*

14. How likely are you to join a new community through an online platform?
34 responses

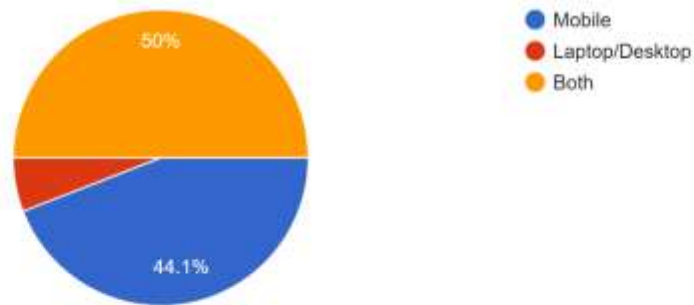


According to the results, over 44% of the people are "very likely" and nearly 47% of people are "likely" to join a new community using an online platform. So it is evident that most users of Smart Connects would be eager to join new communities using an online platform and so it makes this a crucial aspect of Smart Connects.

Figure 29*Question No. 15*

15. What device do you usually use to access online platforms?

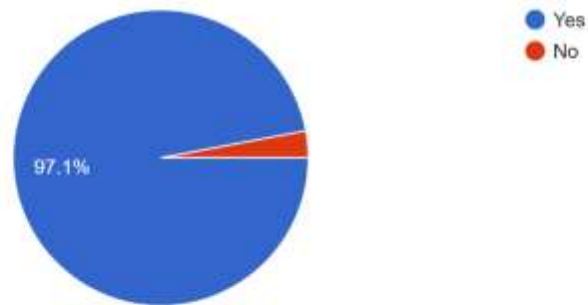
34 responses



This result shows that exactly 50% of people use both devices to access the online platforms whereas 44.1% of people use only mobile device and very few users use only the desktop. So the design of the platform Smart Connects must be responsive so that people can access it on both the platforms.

Figure 30*Question No. 16*

16. Are you concerned about your personal data privacy?
34 responses

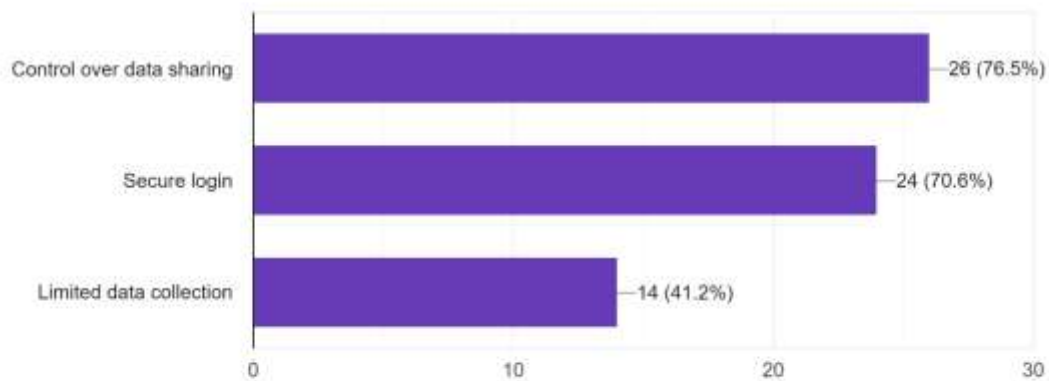


According to the survey, almost all of the users (97.1%) consider the privacy of their personal data as an issue whereas only few are not interested. Thus, privacy and security of user data is an important aspect which needs to be addressed while developing Smart Connects.

Figure 31*Question No. 17*

17. What privacy features would you prefer?

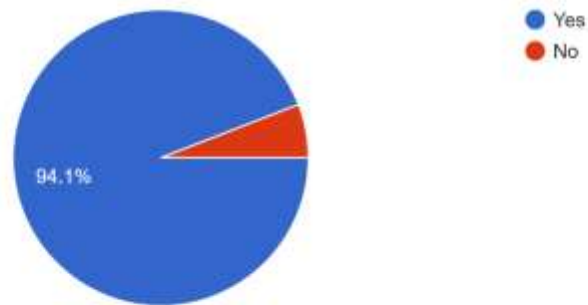
34 responses



According to the survey, majority of users (76.5%) favors that they have control over data sharing. Next majority users are concerned with the security of login process (70.6%). The other 41.2% of users also want limited collection of personal data. It reflects that the users are concerned with control and security of their personal data on the Smart Connects platform.

Figure 32*Question No. 18*

18. Would you prefer personalized recommendations based on your interests?
34 responses



According to the survey, 94.1% of users prefer personalised recommendation based on their interest. This also shows the high interest in personalised services by the users for Smart Connects platform.

Figure 33*Question No. 19***19. What features or improvements would you like to see in Smart Connects to enhance community discovery and user experience?**

34 responses

Good information flow and presentation

notification for new events, rental supplies for events like hikes etc

Must be effective and focused on ease to connect individuals having hard time to find relevant grps

Better community suggestions based on real interests and goals

Location, age, gender and interest based personalized recommendation

Here are several features and improvements that could transform Smart Connect into a more community-centric experience.

- 1.Community "Workflow" Gallery
- 2.Shared "Workspaces" for Collaboration
- 3.Intelligent "Context-Aware" Discovery
- 4.Gamified Community Contributions

As you can see from the comments users were looking for things such as recommendations tailored to them, locations based discovery, and stronger search functions, allowing for easily discovering interesting communities. Several other items that came up in comments include live updates, notifications for events and a user friendly interface for easier navigation. Privacy and control, a secure and verified system and ways for people to communicate were also highlighted. Overall people were seeking an intelligent, and user-friendly and dependable platform to discover and connect to communities.

3.3 System Architecture

The system architecture of Smart Connects represents the overall working flow of the platform from user interaction to data processing and output. The system follows a web-based architecture where users interact through the frontend interface, and all requests are processed through the backend and database.

The process begins when a user accesses the platform through a web browser using a mobile or desktop device. The frontend, built using Next.js, provides the user interface where users can register, log in, search for communities, and view events. User actions such as searching or selecting a community are sent as requests to the backend server.

The backend, developed using Node.js and Express.js, processes these requests and handles the application logic. It retrieves relevant data from the PostgreSQL database, which stores user information, community details, and event data. The backend ensures that the correct data is fetched and sent back to the frontend.

The database acts as the central storage system, maintaining structured records of users, communities, and events. It supports efficient data retrieval and ensures that the system remains organized and scalable.

Finally, the processed data is displayed to the user through the frontend interface in a clear and structured format. This architecture ensures smooth communication between components and provides a reliable and user-friendly platform for discovering communities and events.

Chapter 4: CONCLUSION

Smart Connects provides a well-designed application to the users that are unable to find and join online communities efficiently. The survey clearly illustrates the scattered sources and locations that users were previously relying on without the ability to efficiently find their community without much frustration. This platform offers a centralized location where communities are structured and arranged properly so that users will be able to find the communities they need more easily, raising discoverability and user interaction through structured information and event postings. The survey validates the user need for a central location that brings together communities and their associated events. It will increase user interaction with community organizers with instantaneous access to required information and participation in community events due to the reduction in chance users might overlook certain details and information. User-friendly is key in this application as this will enhance navigation and user interaction.

Besides user navigation and friendliness in the application, user demand for simplicity, convenience and security with specific recommendations and intuitive interaction have been verified by the survey. Also clear from the survey that a real-time approach with curated recommendations is very important for user experience. The users will be concerned regarding data protection and privacy, this will be an ongoing factor during the system design. Although the developed system appears to be very capable it requires additional development concerning UI improvement, security and system scalability. Additional features could also be incorporated to increase its usability further.

4.1 Project Achievements

The Smart Connects project was set off to a solid ground with research and analysis. Users' needs are successfully defined through the survey results which also determines the core functionalities and structure of the system. Therefore, users' behaviors, problems and desires are identified to guide the design. In this process, a deep insight into several critical problems including "information lost" and "how to find the relevant communities" are revealed that guide the architecture and functionalities of the system and confirm that the system is built on users' real needs rather than assumptions.

The research process of the project produced a system architecture. The basic architecture supporting user interaction, information retrieval and processing is defined. Essential functionalities including community listing, events information and personalized suggestions are determined. The architecture helps to structure the system development and integration with other modules. It provides a foundation for extensibility of the system.

4.2 Research Limitations

Although this research forms a good foundation for future work, there are a number of issues that would benefit from further investigation. Currently, this research covers the functionalities and requirements of the system, while there is less analysis of technical factors like performance tuning and processing large amounts of data. The system has not been tested on a real-world large scale system and this would be useful in assessment of the system's performance. We can focus on further enhancement of recommending techniques and also scalability of the system. Future research will aim at providing the better performance of the system with advanced modern technologies such as AI based analytics.

A significant drawback in this work is the absence of an analysis on the long-term behavior of the users. In the future, we may involve continuous feedback and testing from users, in order to improvise the system with respect to user behavior. Features can be improvise on the real usage. The system can be improvised to include mobile integration and enhanced security features to develop a better and a stable system. Also more research has to be done to improvise user retention and engagement.

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APPENDIX

Appendix 1 – PPF



Project Proposal Form

Student Name: Mohit Bhandari

Supervisor: Shambhu Gautam, Bikul Raj Koirala, Rajan Raj Pant

Student No: NP069596

Intake: NP3F2509IT

Email Address: mohit.bhandari@study.lbef.edu.np

Programme Name: B. Sc. (Hons.) Information Technology

Title of Project: Smart Connects: AI powered platform for connecting individuals to communities

Introduction

Clubs, non-governmental organizations (NGOs), and social communities play an important role in supporting social interaction, volunteering, learning, and personal development. These communities operate across different digital platforms, including social media and independent websites. However, information related to these communities is widely scattered, which makes it difficult for individuals to easily discover communities that match their interests, location, and personal goals. As a result, users often spend significant time searching across multiple platforms to find suitable communities, leading to inefficiency and frustration.

Although many clubs and organizations are actively carrying out valuable work, they are not always visible to people in a particular city or area. Research on online platforms and community systems indicates that limited visibility and poor discoverability are common challenges for smaller and local communities, even when they are active and impactful (Valencia & Wibowo, 2025). This lack of structured and centralized information reduces opportunities for individuals to connect with relevant communities and limits the ability of community organizations to reach appropriate audiences (Oghenemaro Anuyah, 2023).

Smart Connects is proposed as a centralized web-based platform that organizes information about clubs, NGOs, and communities in one place. The system aims to improve community visibility and simplify the discovery process by allowing users to search and browse communities based on category, interests, and location. By offering structured listings and basic keyword-based matching, the platform seeks to strengthen connections between individuals and communities while supporting community growth and participation.

Problem Statement

Despite the rapid growth of online clubs, NGOs, and social communities, individuals continue to face difficulties in identifying communities that are relevant to their interests and local area. Prior research on digital platforms shows that users often rely on manual searching across multiple websites and social media platforms, which is time-consuming and leads to information overload (Gavshon & Gorur, 2019). This fragmented environment reduces the efficiency of community discovery and makes it difficult for users to identify active and suitable communities in a structured manner.

Existing platforms commonly prioritize popular or trending groups rather than focusing on communities that match individual user preferences or local needs. Studies on online platform design and community engagement indicate that this popularity-driven approach limits the visibility of smaller and local communities, even when they provide meaningful social value (Dodds et al., 2023). As a result, users may fail to connect with appropriate communities, while community organizations struggle to reach their intended audience (Dodds et al., 2023).

Furthermore, many communities and NGOs lack access to simple tools for managing their online presence, promoting events, and monitoring user engagement. Research on digital community management highlights that the absence of centralized systems for visibility and engagement tracking reduces community growth and participation (Marketing, 2023). Without a structured platform to support both users and communities, the process of connecting individuals to relevant groups remains inefficient, unorganized, and dependent on manual search practices.

Aim

To design and develop a centralized web-based platform that helps individuals easily discover and connect with relevant clubs, NGOs, and communities based on interests and location.

Objectives

- To design and develop a web-based application for listing and discovering clubs, NGOs, and community groups.
- To enable users to search and filter communities using categories, interests, and location-based keywords.
- To provide basic keyword-based matching to support personalized community discovery within the web platform.
- To improve community visibility by offering profile management and event promotion features through the web application.

Literature Review

Online communities and social groups have been widely studied for their role in improving social interaction, knowledge sharing, and personal development. Research shows that clubs, NGOs, and community-based groups help individuals build social connections and a sense of belonging, especially in digital environments (Hwang & Foote, 2021). However, several studies highlight that users often face difficulties in finding relevant communities due to information being scattered across multiple platforms, leading to inefficient discovery and reduced participation.

Previous research on digital platforms emphasizes the importance of centralized information systems for improving accessibility and visibility. Studies suggest that when community information is organized in a structured and searchable manner, users are more likely to discover suitable groups that match their interests (Bonina et al., 2021). Keyword-based search and category filtering have been identified as effective techniques for improving content discovery without requiring complex recommendation mechanisms (Javed et al., 2021). Such approaches are commonly used in web-based platforms to enhance user experience while keeping system design simple and practical.

Additionally, research on community management systems indicates that smaller and local communities often struggle with visibility despite being active and valuable. Limited online presence and lack of proper tools for event promotion and engagement tracking reduce their ability to reach the right audience (Studio Berlin, 2025). Literature supports the development of web-based platforms that provide profile management, event listings, and basic engagement features to support community growth. These findings justify the need for a centralized web application that improves both community discoverability and organizational visibility through simple and user-friendly design.

Deliverables

- **Web-Based Application:** A functional web application for discovering clubs, NGOs, and communities.
- **Community Listing Module:** Structured community profiles with basic information such as category, contacts and location.
- **Search and Filtering Functionality:** Keyword-based search and category filters to support easy community discovery.
- **Community Management Interface:** A simple interface for communities to manage profiles and post activities.
- **Database Design:** Logical database structure to store users, communities, and related information.
- **Final Project Documentation:** Complete documentation covering analysis, design, implementation, and evaluation.

References

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Appendix 2 - PSF



STUDENT ID : NP069596
INTAKE : NP3F2509IT
STUDENT NAME : Mohit Bhandari

A. Project Title

"Smart Connects: AI powered platform for connecting individuals to Communities"

B. Brief description of project background. (i.e. problem context, rationale, description of problem area, nature of challenge)

Problem Context

Many students and individuals struggle to find suitable clubs and communities based on their interests, skills, and location. Existing platforms such as Facebook Groups and Meetup promote popular or trending communities rather than personalized matches. Smaller clubs remain hidden due to low visibility and limited digital structure (Centola, 2021).

When people migrate to a new city for study or work, they often face difficulty adjusting to a new environment. They lack awareness about local communities aligned with their interests. This affects social integration and overall quality of life. A structured discovery system can help newcomers connect with relevant groups based on hobbies, profession, language, culture, or skill level (Ellison, Steinfield, & Lampe, 2021).

The platform also supports different age groups. Senior citizens can form age-based communities or interest clubs within their locality. This improves social interaction, reduces isolation, and strengthens community bonding (Phua, Jin, & Kim, 2021).

In universities and cities, multiple clubs operate independently without a centralized searchable system. Users spend excessive time browsing scattered social media pages and unstructured listings. There is no efficient keyword-based filtering mechanism that allows users to discover clubs based on specific interests, city, category, or expertise level.

The challenge is to design a centralized web platform that provides structured listings, intelligent keyword matching, and limited AI assisted recommendations. The system will prioritize usability, search efficiency, and structured data filtering instead of heavy chatbot automation (Burke, Kraut, & Marlow, Social Capital on Facebook: Differentiating Uses and Users, 2021).

Rationale

According to the problems identified, "Smart Connects: AI Powered Platform for Connecting Individuals to Communities" provides a centralized digital system that helps users discover relevant clubs and communities based on interests, location, and preferences. The platform replaces scattered social media searches with structured keyword-based discovery. It prioritizes personalized matching over trending promotion and increases visibility for small or inactive clubs (Adomavicius & Tuzhilin, 2025).

The platform supports students and professionals who move to new cities by enabling faster social integration. Users can search communities aligned with hobbies, academics, career goals, or cultural background. Senior citizens and other age groups can create and join interest based or age specific clubs to strengthen engagement and improve quality of life. Structured filtering and limited AI driven recommendations ensure accurate discovery and inclusive participation (Resnick & Varian, 2021).

Tangible Benefits:

- Faster discovery of relevant clubs
- Structured filtering by category and location

- Increased visibility for small and inactive clubs
- Encourages local interaction and real world community engagement

Intangible Benefits

- Stronger community engagement.
- Improved student participation
- Increased collaboration across interest groups
- Digital transformation of local club management

Nature of Challenge

Developing Smart Connects involves multiple technical and operational challenges. The system requires a scalable database architecture to manage growing users and club records without performance issues. Accurate keyword-based recommendation logic must be implemented to ensure relevant search results. A clean and intuitive user interface is essential to provide smooth navigation for all age groups. The platform must also maintain data authenticity through proper admin verification and implement secure authentication to protect user information (Lampe, Ellison, & Steinfield, 2022). In addition, AI integration must remain limited and controlled within a small scope to avoid unnecessary system complexity while still enhancing search accuracy.

C. Brief description of project objectives. (I.e. scope of proposal and deliverables)

Deliverables

"Smart Connects: AI Powered Platform for Connecting Individuals to Communities" aims to design and develop a centralized web platform for structured club listing and community discovery. The system will use keyword-based matching to help users find relevant clubs in their surrounding area according to interests and location. Club organizers will register and manage their clubs through an admin approval process to maintain authenticity and data

quality. The platform will feature a responsive design to ensure smooth and consistent access across desktop and mobile devices.

Admin Module

- Add, update, and delete club listings
- Approve club registration requests
- Manage categories and keywords
- Monitor user activity.

User Module

- Register and login securely
- Search clubs using keywords
- Filter by city and category
- Receive basic personalized suggestions

Club Organizer Module

- Submit club registration request
- Update club profile details
- Add events and announcements

D. Brief description of the resources needed by the proposal. (i.e. hardware, software, access to information/expertise, user involvement, etc.)

Hardware

The minimum requirements for hardware for the personal computer to successfully carry out and meet the objectives of the system are as follows:

- Processor – Pentium 4 2.66A (2.66 GHz)
- Random Access Memory (RAM) – 1GB
- Keyboard & Mouse
- Router (RJ45 / Wireless Fidelity (Wi-Fi))

Software

The minimum software requirements for the development and execution of the project are as follows:

Database Management System (DBMS)

- PostgreSQL

Server-Side Scripting, Web Server

- Node + Express.js
- Nginx Server
- Code editor (Visual Studio)

Documentation and Planning

- Microsoft Project 2016
- Microsoft Word 2016

Access to Information / Expertise

The consultation is required from experts who have expertise in the field of developing web applications. Information that is required during the project can be gathered from the field survey.

User Involvement

The system will be developed by considering the needs of diverse users, including students, professionals, senior citizens, and local community members. Individuals who move to new cities for study or work will be primary beneficiaries, as the platform will help them find relevant clubs and integrate into their surroundings. Club organizers will also play an important role, as they will register and manage their communities through the system (Burke & Kraut, The Relationship Between Facebook Use and Well-Being, 2023).

Feedback will be collected from students, local residents, and club organizers during the development process. Their suggestions will be used to refine features such as search accuracy, filtering options, and user interface design. Selected users will participate in usability testing to evaluate navigation, response time, and overall user experience before final deployment.

E. Academic research being carried out and other information, and techniques being learned. (i.e. what are the names of books you are going to read / data sets you are going to use)

The following academic books, peer-reviewed journals, and online research resources will be referenced to the study and development of this project to strengthen the theoretical understanding and technical implementation of deep learning based image detection.

Books

To carry out the deliverables, the preliminary list of books and web pages I will study are as follows.

Books

- Name: Nodejs and PostgreSQL Web Development (4th Edition)
 Author: Luke Welling, Laura Thomson
 Publisher: Addison-Wesley Professional
- Name: Learning Node.js

Author: David Sklar

Publisher: O'Reilly Media

- Name: Murach's PostgreSQL and SQL for Developers, 2nd Edition

Author: Joel Murach

Publisher: Murach

- Name: Node.js: A beginner's guide

Author: Vikram Vaswani

Publisher: McGraw-Hill

- Name: The Web Server Book

Author: Jonathan Magid, Paul Jones, R. Douglas Matthews

Publisher: Ventana

Online Resources

- Ivan Mustakerov, Daniela Borrisova, "A conceptual approach for development of educational web-based e-testing system", Expert System with applications, Volume 38
- Tolga Kahraman, Seref Sagiroglu, Ilhami Colak," Development of adaptive and intelligent web-based educational systems", IEEE
- Lora Aroyo, Darina Dicheva, "The new challenges for e-learning: The educational Semantic Web", International Forum of educational technology and society, Volume 4

- Natalya Prokofyeva, Victoria Boltunova, "Analysis and Practical Application of Node.js in Development of Web Information Systems", *Procedia Computer Science*, Volume 104
- Piero Fraternali, "Tools and approaches for developing data-intensive web applications: A Survey", *ACM computing surveys*, Volume 31

F. Brief description of the development plan for the proposed project. (i.e. which software methodology and why, the major areas of functions to be developed, and the order in which developed)

System Development Methodology

For the development of "Smart Connects: AI Powered Platform for Connecting Individuals to Communities," an Agile methodology with a prototyping approach will be used. Agile is selected because the project requires continuous feedback, iterative improvement, and flexibility during development. Since user experience and search accuracy are critical, early prototypes will be created to gather feedback and refine system features before full implementation. This approach reduces risk, improves usability, and ensures that the final system meets user requirements effectively (Ricci, Rokach, & Shapira, 2025).

Major Functional Areas and Development Order:

The development process will follow a structured sequence. First, requirement analysis and system design will be completed, including database schema and system architecture. Second, the authentication module will be developed to manage secure user and organizer registration and login. Third, the club listing and management module will be implemented, allowing organizers to submit clubs and admins to approve them (Shania, Raharjo, & Fitriani, 2023).

After that, the search and keyword-based discovery system will be developed to enable structured filtering by interest and location. Next, the personalized recommendation logic with limited AI integration will be implemented. Finally, the user dashboard, testing, system integration, and deployment phases will be completed to ensure performance, security, and usability.

G. Brief description of the evaluation and test plan for the proposed project. (i.e. what are the success criteria and how will be evaluated and implementation will be tested, indicate the estimated size of the demonstration/test database)

Success Criteria

The main goal of “Smart Connects: AI Powered Platform for Connecting Individuals to Communities” is to ensure accurate club discovery, efficient keyword-based search, secure registration, and smooth user interaction. The system will be considered successful if users are able to register, search clubs based on interests and location, and interact with verified community listings without technical errors. Response time should remain fast and search results should match user keywords accurately.

Unit Testing

Each module of the system will be tested separately to ensure correct functionality. This includes user registration and login, club registration, admin approval process, search and filtering module, and dashboard features. For example, in the registration module, validation rules such as minimum password length and required fields will be tested. Each unit must function correctly before integration into the full system (Putnam, 2020).

Integration Testing

After unit testing, different modules will be combined and tested together. For example, when a club organizer submits a new club and the admin approves it, the club should automatically appear in the public search results. The connection between frontend,

backend, and database will be tested to ensure accurate data retrieval and update operations. Keyword search results will be verified against stored database records to confirm correct filtering.

Usability Testing

Usability testing will be conducted with a group of students and community members. Participants will perform tasks such as searching for clubs, registering, and saving interests. The graphical user interface, navigation clarity, feedback messages, and response time will be evaluated. User feedback will be collected to improve design simplicity and overall user experience.

Estimated Demonstration Database

The demonstration database will contain approximately 20 to 50 club records across different categories and locations. It will also include around 50 to 100 user accounts for testing purposes. Each club profile will contain structured keyword tags to support accurate search filtering and recommendation validation during system testing.

References

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Appendix 3 - Ethic Form

Office Record Date Received: Received by whom:	Receipt - Fast-Track Ethical Approval Student name: Student number: Received by: Date:
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APU / APIIT FAST-TRACK ETHICAL APPROVAL FORM (STUDENTS)
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Tick one box (level of study): ☐ POSTGRADUATE (PhD / MPhil / Masters) ☐ UNDERGRADUATE (Bachelors degree) ☐ FOUNDATION / DIPLOMA / Other categories	Tick one box (purpose of approval): ☐ Thesis / Dissertation / FYP project ☐ Module assignment ☐ Other: _____
Title of Programme on which enrolled: <u>B.Sc IT (Hons)</u>	
Tick one box: ☐ Full-Time Study or ☐ Part-Time Study	
Title of project / assignment: <u>Smart Connects: AI Powered Platform for Connecting Individuals to Communities</u>	
Name of student researcher: <u>Mohit Bhandari</u>	
Name of supervisor / lecturer: <u>Dr. Laxman Mandal</u>	

Student Researchers- please note that certain professional organisations have ethical guidelines that you may need to consult when completing this form.

Supervisors/Module Lecturers - please seek guidance from the Chair of the APU Research Ethics Committee if you are uncertain about any ethical issue arising from this application.

		YES	NO	N/A
1	Will you describe the main procedures to participants in advance, so that they are informed about what to expect?	✓		
2	Will you tell participants that their participation is voluntary?	✓		
3	Will you obtain written consent for participation?	✓		
4	If the research is observational, will you ask participants for their consent to being observed?	✓		
5	Will you tell participants that they may withdraw from the research at any time and for any reason?	✓		
6	With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?	✓		
7	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	✓		
8	Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?	✓		

If you have ticked **No** to any of Q1-8 you should complete the full Ethics Approval Form.

		YES	NO	N/A
9	Will your project/assignment deliberately mislead participants in any way?		✓	
10	Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort?		✓	
11	Is the nature of the research such that contentious or sensitive issues might be involved?		✓	

If you have ticked **Yes** to 9, 10 or 11 you should complete the full Ethics Approval Form. In relation to question 10 this should include details of what you will tell participants to do if they should experience any problems (e.g. who they can contact for help). You may also need to consider risk assessment issues.

		YES	NO	N/A
12	Does your project/assignment involve work with animals?		✓	
13	Do participants fall into any of the following special groups? Note that you may also need to obtain satisfactory clearance from the relevant authorities		✓	
	Children (under 18 years of age)			
	People with communication or learning difficulties			
	Patients			
	People in custody			
	People who could be regarded as vulnerable			
	People engaged in illegal activities (eg drug taking)			
14	Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?		✓	

If you have ticked **Yes** to 12, 13 or 14 you should complete the full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.

STUDENT RESEARCHER

Provide in the boxes below (plus any other appended details) information required in support of your application, THEN SIGN THE FORM.

Please Tick Boxes

I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee.	✓
--	---

Give a brief description of participants and procedure (methods, tests used etc) in up to 150 words.

Participants will include university students and young aged 18 and above who are interested in joining clubs, NGOs, or community groups. Additional Participants will include representatives or administrators of registered clubs, NGOs and social communities, who will create and manage organizational profiles on the platform.

The study will use a prototype AI assisted web platform for community discovery. Participants will be asked to register, complete an interest based onboarding form, explore community listings, and interact with the AI chat bot for recommendations. Community representatives will test profile creation, event posting, and analytics features. Data will be collected through system usage logs, short online questionnaires, and structured feedback interviews. No sensitive personal data will be collected beyond basic profile details. The study evaluates usability, recommendation accuracy, and overall user satisfaction with the platform.

I also confirm that: i) All key documents e.g. consent form, information sheet, questionnaire/interview are appended to this application.	
Or ii) Any key documents e.g. consent form, information sheet, questionnaire/interview schedules which need to be finalised following initial investigations will be submitted for approval by the project/assignment supervisor/module lecturer before they are used in primary data collection.	

E-signature: Mohit Print Name: Mohit Bhandari Date: Feb 23, 2026
 (Student Researcher)

Please note that any variation to that contained within this document that in any way affects ethical issues of the stated research requires the appending of new ethical details. New ethical consent may need to be sought.

The completed form (and any attachments) should be submitted for consideration by your Supervisor/Module Lecturer

**SUPERVISOR/MODULE LECTURER
PLEASE CONFIRM THE FOLLOWING:**

Please Tick Box

I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee	☒
i) I have checked and approved the key documents required for this proposal (e.g. consent form, information sheet, questionnaire, interview schedule) Or ii) I have checked and approved draft documents required for this proposal which provide a basis for the preliminary investigations which will inform the main research study. I have informed the student researcher that finalised and additional documents (e.g. consent form, information sheet, questionnaire, interview schedule) must be submitted for approval by me before they are used for primary data collection.	☒

SUPERVISOR AND SECOND ACADEMIC SIGNATORY

STATEMENT OF ETHICAL APPROVAL (please delete as appropriate)

- 1) THIS PROJECT/ASSIGNMENT HAS BEEN CONSIDERED USING AGREED APU/SU PROCEDURES AND IS NOW APPROVED
- 2) THIS PROJECT/ASSIGNMENT HAS BEEN APPROVED IN PRINCIPLE AS INVOLVING NO SIGNIFICANT ETHICAL IMPLICATIONS, BUT FINAL APPROVAL FOR DATA COLLECTION IS SUBJECT TO THE SUBMISSION OF KEY DOCUMENTS FOR APPROVAL BY SUPERVISOR (see Appendix A)

E-signature: Dr. Laxman Mandal Print Name: Dr. Laxman Mandal Date: February 23, 2026
 (Supervisor/Lecturer)

E-signature: Edward Basnet Print Name: Edward Basnet Date: Feb. 25 2026
 (Second Academic Signatory)

Office Record	Receipt – Appendix A (Fast-Track Ethics Form)
Date Received:	Student name:
Received by whom:	Student number:
	Received by:
	Date:

APPENDIX A
AUTHORISATION FOR USE OF KEY DOCUMENTS

Completion of Appendix A is required when for good reasons key documents are not available when a fast track application is approved by the supervisor/module lecturer and second academic signatory.

I have now checked and approved all the key documents associated with this proposal e.g. consent form, information sheet, questionnaire, interview schedule

Title of project/assignment: Smart Connects : AI Powered Platform for
Connecting Individuals to Communities

Name of student researcher: Mohit Bhandari

Student ID: NP069596 Intake: NP3F2509IT

E-signature:  Print Name: Dr. Laxman Mandal Date: February 23, 2026
(Supervisor/Lecturer)

Appendix 4 - Meeting Log Sheet



Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: <u>Mohit Bhandari</u>	Date: <u>23/02/2021</u>	Meeting No: <u>1</u>
Project Title: <u>Smart Connects: AI Powered for Connecting Individuals to communities</u>		Insk: <u>NP3F2509II</u>
Supervisor's Name: <u>Dr. Laxman Mandal</u>		Signature: <u>[Signature]</u>
Items for discussion (noted by student <u>before</u> mandatory supervisory meeting):		
1. Confirmation of project title.		
2. Clarification of system features		
3. Discussion of AI recommendation approach		
4.		
5.		
Record of discussion (noted by student <u>during</u> mandatory supervisory meeting):		
1. Supervisor approved the proposed project title.		
2. Agreed the system will focus on community discovery.		
3. Make the project simple that can be finished in the timeframe.		
4.		
5.		
Action List (to be attempted or completed by student by the <u>next</u> mandatory supervisory meeting):		
1. Read the papers for more clarification		
2. Make the project Frame.		
3.		
4.		
5.		

STUDENT COPY

Project Log Sheet – Supervisory Session

Notes on use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: Mohit Bhandari	Date: 18/03/2026	Meeting No: 02
Project Title: Smart Connects! AI Powered Platform for connecting individuals to communities. Intake: NP3F2509IT		
Supervisor's Name: Dr. Lakman Mandal	Signature: 	
Items for discussion (noted by student <u>before</u> mandatory supervisory meeting):		
1. Discussion on the drafted Chapter 1: Introduction 2. Review of project timeline and planned development phases. 3. Discussion on UI/UX design and tools for the platform interface.		
Record of discussion (noted by student <u>during</u> mandatory supervisory meeting):		
1. Supervisor reviewed the drafted Chapter 1 documentation. 2. Suggested refining the problem background and objectives. 3. Advised to keep the project scope realistic & achievable. 4. Advised to continue on the literature review and working on UI/UX design.		
Action List (to be attempted or completed by student by the <u>next</u> mandatory supervisory meeting):		
1. Improve Chapter 1 based on the supervisor feedback. 2. Prepare initial UI/UX design for the platform. 3. Continue working on the documentation.		

STUDENT COPY



Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which questions of he or she needs to ask the supervisor and what progress has been made (if any) since the last session and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic court.
6. A copy is returned by the student to be filed in the project file.
7. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
8. The log sheet is an important document for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: Mohit Bhandari	Date: 06/04/2026	Meeting No: 03
Project Title: Smart Connects: AI powered platform for connecting individuals to communities. Intake: NP3F2509IT		
Supervisor's Name: Dr. Laxman Mandal	Signature:	
Items for discussion (noted by student before mandatory supervisory meeting):		
1. Progress update on Chapter 2 Literature Review. 2. Discussion on Chapter 3 Methodology structure. 3. Discussion on data gathering process for communities and events. 4. Review of the Smart Connects home page UI design. 5. Feedback on overall project development progress.		
Record of discussion (noted by student during mandatory supervisory meeting):		
1. Supervisor reviewed Chapter 2 and suggested improvements. 2. Discussed Chapter 3 and confirmed Agile Methodology. 3. Discussed data gathering process for smart Connects platform. 4. Reviewed the UI design & suggested improvements in layout. 5. Supervisor advised to continue development of system design & architecture.		
Action List (to be attempted or completed by student by the next mandatory supervisory meeting):		
1. Revise Chapter 2 based on supervisor feedback. 2. Complete Chapter 3 methodology section. 3. Improve UI design & create additional pages. 4. Start making system architecture & collecting data.		

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Project Log Sheet – Supervisory Session

Notes on use of the project log sheet:

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session and rating these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: Mohit Bhandari	Date: 20/04/2026	Meeting No: 4
Project Title: Smart Connects: AI Powered for connecting Individuals to Communities		Intake: NP3F250921
Supervisor's Name: Dr. Laxman Mandal	Signature:	
Items for discussion (noted by student before mandatory supervisory meeting):		
1. Completion of Chapter 2 Literature Review 2. Progress update on Chapter 3 Methodology structure. 3. Review of Website UI design and layout 4. Finalizing the survey questions for data collection.		
Record of discussion (noted by student during mandatory supervisory meeting):		
1. Supervisor reviewed Chapter 3 and suggested correction of system Architecture. 2. Reviewed UI design and suggested layout improvements. 3. Approved survey questions for data collection. 4. Discussed on Methodology structure and clarity. 5. Reviewed Chapter 4 and suggested refinements.		
Action List (to be attempted or completed by student by the next mandatory supervisory meeting):		
1. Improve website UI design and add more pages. 2. Distribute survey form and start collecting the data. 3. Complete Chapter 3 Methodology. 4. Refine Chapter 4 Conclusion.		

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Project Log Sheet – Supervisory Session

Notes in use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project's SIX mandatory supervisory sessions.
2. The student should prepare for the supervisory sessions by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisory session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: <u>Mohit Bhandari</u>	Date: <u>27/04/2026</u>	Meeting No: <u>9</u>
Project Title: <u>Smart Connects: AI Powered for Connecting Individuals to Communities</u>		Intake: <u>NP3F2509 IT</u>
Supervisor's Name: <u>Dr. Laxman Mandal</u>	Signature: <u>[Signature]</u>	
Items for discussion (noted by student <u>before</u> mandatory supervisory meeting):		
1. Review of completed chapters. 2. Reviewing final documentation and report formatting. 3. Discuss Preparation for final submission. 4. Reviewing the Gantt Chart. 5.		
Record of discussion (noted by student <u>during</u> mandatory supervisory meeting):		
1. Supervisor reviewed completed chapters and suggested minor corrections. 2. Checked final document and confirmed formatting consistency. 3. Discussed readiness for final submission. 4. Reviewed the Gantt Chart and confirmed project timeline. 5.		
Action List (to be attempted or completed by student by the <u>next</u> mandatory supervisory meeting):		
1. Apply final corrections to all chapters. 2. Finalize report formatting and documentation. 3. Submit final report and project files. 4. 5.		

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